

CONSOLIDATED VERSION



**Semiconductor devices –
Part 16-3: Microwave integrated circuits – Frequency converters**



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SEMICONDUCTOR DEVICES –

**Part 16-3: Microwave integrated circuits –
Frequency converters**

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In this Redline version, a vertical line in the margin shows where the technical content is modified by amendments 1 and 2. Additions are in green text, deletions are in strikethrough red text. A separate Final version with all changes accepted is available in this publication.

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SEMICONDUCTOR DEVICES –

Part 16-3: Microwave integrated circuits – Frequency converters

1 Scope

This part of IEC 60747 provides new measuring methods, terminology and letter symbols, as well as essential ratings and characteristics for integrated circuit microwave frequency converters.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

~~IEC 60050-702:1992, International Electrotechnical Vocabulary – Chapter 702: Oscillations, signals and related devices~~

~~IEC 60617-12, Graphical symbols for diagrams – Part 12: Binary logic elements~~

~~IEC 60617-13, Graphical symbols for diagrams – Part 13: Analogue elements~~

~~IEC 60747-1:1983, Semiconductor devices – Discrete devices and integrated circuits – Part 1: General~~

~~IEC 60747-16-1:2001, Semiconductor devices – Part 16-1: Microwave integrated circuits – Amplifiers –
Amendment 1 (2007)¹~~

IEC 60050-702, International Electrotechnical Vocabulary – Chapter 702: Oscillations, signals and related devices (available at < <http://www.electropedia.org/> >)

IEC 60617, Graphical symbols for diagrams (available at < <http://std.iec.ch/iec60617> >)

IEC 60747-1:2006, Semiconductor devices – Part 1: General
IEC 60747-1:2006/AMD 1:2010

IEC 60748-2:1997, Semiconductor devices – Integrated circuits – Part 2: Digital integrated circuits

IEC 60748-3, Semiconductor devices – Integrated circuits – Part 3: Analogue integrated circuits

IEC 60748-4, Semiconductor devices – Integrated circuits – Part 4: Interface integrated circuits

¹ ~~There exists a consolidated edition 1.1 published in 2007, including the base publication (2001) and its Amendment 1 (2007).~~

IEC 61340-5-1:2007, *Electrostatics – Part 5-1: Protection of electronic devices from electrostatic phenomena – General requirements*

IEC/TR 61340-5-2:2007, *Electrostatics – Part 5-2: Protection of electronic devices from electrostatic phenomena – User guide*

3 Terms and definitions

For the purpose of this part of IEC 60747, the following terms and definitions apply:

3.1

conversion gain, G_c

ratio of the desired converted output power to the input power

NOTE Usually, the conversion gain is expressed in decibels.

3.2

conversion gain flatness, ΔG_c

difference between the maximum and the minimum conversion gain for a specified input power in a specified frequency range

3.3

LO/Rf isolation, $P_{LO}/P_{LO(RF)}$

ratio of the incident local power to the local leakage power at the RF port with the IF port terminated in a specified impedance

3.4

LO/IF isolation, $P_{LO}/P_{LO(IF)}$

ratio of the incident local power to the local leakage power at the IF port with the RF port terminated in a specified impedance

3.5

RF/IF isolation, $P_{RF}/P_{RF(IF)}$

ratio of the incident RF power to the RF feedthrough power at the IF port for a specified local power

NOTE Usually, the RF/IF isolation is applied to the down-converter.

3.6

IF/Rf isolation, $P_{IF}/P_{IF(RF)}$

ratio of the incident IF power to the IF feedthrough power at the RF port for a specified local power

NOTE Usually, the IF/Rf isolation is applied to the up-converter.

3.7

image rejection, $P_o/P_{o(im)}$

ratio of the output power when the RF signal is applied, to the output power when the image signal is applied

NOTE Usually, the image rejection is applied to the down-converter.

3.8

sideband suppression, $P_o/P_{o(U)}$

ratio of the output power of the desired sideband to the output power of the undesired sideband

NOTE Usually, the sideband suppression is applied to the up-converter.

3.9

LO port return loss, $L_{\text{ret(LO)}}$

ratio of the specified incident power at the LO port to the reflected power at the LO port, with the RF port and the IF port terminated in each specified impedance

3.10

RF port return loss, $L_{\text{ret(RF)}}$

ratio of the incident power at the RF port to the reflected power at the RF port for a specified local power, with the IF port terminated in a specified impedance

3.11

IF port return loss, $L_{\text{ret(IF)}}$

ratio of the incident power at the IF port to the reflected power at the IF port for a specified local power, with the RF port terminated in a specified impedance

3.12

output power

P_o

~~see IEC 60747-16-2, 3.3²~~ RF power measured at the output port

[SOURCE: IEC 60747-4:2007/AMD1:2017, 7.2.2]

3.13

output power at 1-dB conversion compression, $P_{o(1dB)}$

output power where the conversion gain decreases by 1 dB compared with the linear conversion gain

3.14

noise figure, F

~~see IEC 60747-1 Chapter IV, 5.4.4~~ see 702-08-57 of IEC 60050-702

NOTE The term "noise figure" expresses "noise factor" in decibels.

3.15

intermodulation distortion

P_n/P_1

~~ratio of the output power of the n th-order component to the output power of the fundamental component, at a specified input power~~

~~see 3.7 of Amendment 1 of IEC 60747-16-1~~

ratio of the n th order component of the output power to the fundamental component of the output power

NOTE The abbreviation " IMD_n " is in common use for the n th order intermodulation distortion.

[SOURCE: IEC 60747-4:2007/AMD1:2017, 7.2.19]

3.16

output power at the intercept point (for intermodulation products), $P_{n(IP)}$

output power at the intersection between the extrapolated output powers of the fundamental component and the n th order intermodulation components, when the extrapolation is carried out in a diagram showing the output power of the components (in decibels) as a function of the input power (in decibels)

4 Abbreviated terms

The abbreviations used in this part of IEC 60747 are as follows:

² ~~IEC 60747-16-2:2001, Semiconductor devices—Part 16-2: Microwave integrated circuits—Frequency prescalers~~

RF Radio Frequency;

IF Intermediate Frequency;

LO Local Oscillator.

5 Essential ratings and characteristics

5.1 General

This clause gives ratings and characteristics required for specifying integrated circuit microwave frequency converters.

5.1.1 Circuit identification and types

5.1.1.1 Designation and types

The identification of type (device name), the category of circuit and technology applied shall be given.

Microwave frequency converters are divided into two categories:

- type A: down-converter;
- type B: up-converter.

5.1.1.2 General function description

A general description shall be made of the function performed by the integrated circuit microwave frequency converters and the features for the application.

5.1.1.3 Manufacturing technology

The manufacturing technology, for example, semiconductor monolithic integrated circuit, thin film integrated circuit, micro-assembly, shall be stated. This statement shall include details of the semiconductor technologies such as Schottky-barrier diode, MESFET, Si bipolar transistor, HBT.

5.1.1.4 Package identification

The following shall be stated:

- a) chip or packaged form;
- b) IEC and/or national reference number of the outline drawing, or ~~of~~-drawing of non-standard package including terminal numbering;
- c) principal package material, for example, metal, ceramic, plastic;
- d) for chip form: outlines, dimensions, pad sizes, contact material, and recommended contact technologies.

5.1.1.5 Main application

The main application shall be stated if necessary. If the device has restrictive applications, these too shall be stated here.

5.2 Application description

Information on the application of the integrated circuit and its relation to the associated devices shall be given.

5.2.1 Conformance to system and/or interface information

It shall be stated whether the integrated circuit conforms to an application system and/or an interface standard or recommendation.

Detailed information about application systems, equipment and circuits such as VSAT systems, DBS receivers, microwave landing systems shall also be given.

5.2.2 Overall block diagram

A block diagram of the applied systems shall be given if necessary.

5.2.3 Reference data

The most important properties required to permit comparison between derivative types shall be given.

5.2.4 Electrical compatibility

It shall be stated whether the integrated circuit is electrically compatible with other particular integrated circuits or families of integrated circuits, or whether special interfaces are required.

Details shall be given concerning the type of the input and output circuits, for example, input/output impedances, d.c. block, open-drain.

Interchangeability with other devices, if any, shall be given.

5.2.5 Associated devices

If applicable, mention shall be made here of

- devices necessary for correct operation (list with type number, name, and function);
- peripheral devices with direct interfacing (list with type number, name, and function).

5.3 Specification of the function

5.3.1 Detailed block diagram – functional blocks

A detail block diagram or equivalent circuit information of the integrated circuit microwave frequency converters shall be given. The block diagram shall be composed of the following:

- a) functional blocks;
- b) mutual interconnections among the functional blocks;
- c) individual functional units within the functional blocks;
- d) mutual interconnections among the individual functional blocks;
- e) function of each external connection;
- f) interdependence between the separate functional blocks.

The block diagram shall identify the function of each external connection, and, where no ambiguity can arise, it can also show the terminal symbols and/or numbers. If the encapsulation has metallic parts, any connection to them from external terminals shall be indicated. The connections with any associated external electrical elements shall be stated, where necessary.

For the purpose of providing additional information, the complete electrical circuit diagram can be reproduced, though this will not necessarily involve giving indications of the function. Rules governing such diagrams may be obtained from ~~IEC 60617-12 or IEC 60617-13~~ IEC 60617.

5.3.2 Identification and function of terminals

All terminals shall be identified on the block diagram (supply terminals, input or output terminals, input/output terminals).

The terminal functions 1) to 4) shall be indicated, as shown in table 1 below.

Table 1 – Function of terminals

Terminal number	Terminal symbol	1 Terminal designation	2 Function	3 Input/output identification	4 Type of input/output circuits

5.3.2.1 Function 1: Terminal designation

The terminal designation to indicate the terminal function shall be given. Supply terminals, ground terminals, blank terminals (with abbreviation NC) and non-usable terminals (with abbreviation NU) shall be distinguished.

5.3.2.2 Function 2: Function

A brief indication of the terminal function shall be given:

- each function of multi-role terminals, i.e. terminals having multiple functions;
- each function of the integrated circuit selected by mutual pin connections, or programming and/or application of function selection data to the function selection pin such as mode selection pin.

5.3.2.3 Function 3: Input/output identification

Input, output, input/output, and multiples of the input/output terminal shall be distinguished.

5.3.2.4 Function 4: Type of input/output circuits

The type of input and output circuits, for example, input/output impedances, with or without d.c. block, etc., shall be distinguished.

5.3.2.5 Function 5: Type of ground

If the baseplate of the package is used as ground, this shall be stated.

EXAMPLE:

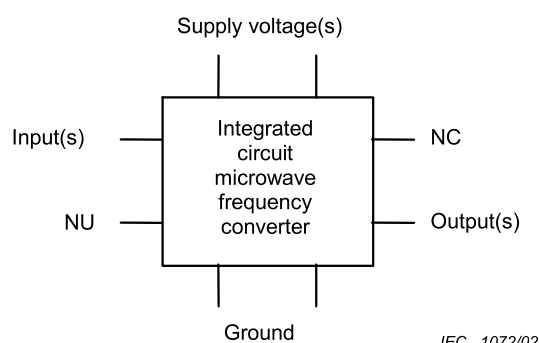


Figure 1 – Electrical terminal symbols

5.3.3 Function description

The function performed by the circuit shall be specified, including the following information:

- basic function;
- relation to external terminals;
- operation mode (e.g., set-up method, preference, etc.);
- interruption handling.

5.3.4 Family related characteristics

All family-specific functional descriptions shall be stated (with reference to IEC 60748-2, IEC 60748-3 and IEC 60748-4).

If ratings, characteristics and function characteristics exist for the family, the relevant part of IEC 60748 shall be used (for example, for microprocessors, see IEC 60748-2, chapter III, section 3).

NOTE For each new device family, specific items shall be added to the relevant part of IEC 60748.

5.4 Limiting values (absolute maximum rating system)

The table giving these values shall specify the following:

- any interdependence of limiting conditions;
- if externally connected and/or attached elements, for example heatsinks, have an influence on the values of the ratings, the ratings shall be prescribed for the integrated circuit with the elements connected and/or attached;
- if limiting values are exceeded for transient overload, the permissible excess and their durations shall be specified;
- where minimum and maximum values differ during programming of the device, this shall be stated;
- all voltages referenced to a specified reference terminal (V_{ss} , GND, etc.);
- in satisfying the following clauses, if maximum and/or minimum values are quoted, the manufacturer shall indicate whether he refers to the absolute magnitude or to the algebraic value of the quantity;
- the ratings given shall cover the operation of the multi-function integrated circuit over the specified range of operating temperatures. Where such ratings are temperature-dependent, such dependence shall be indicated.

5.4.1 Electrical limiting values

Limiting values shall be specified as follows:

Table 2 – Electrical limiting values

Subclause	Parameters	Min.	Max.
5.4.1.1	Ambient or case temperature	±	±
5.4.1.2	Storage temperature	±	±
5.4.1.3	Power supply voltage(s)		±
5.4.1.4	Power supply current(s) (where appropriate)		±
5.4.1.5	Terminal voltage(s) (where appropriate)	±	±
5.4.1.6	Terminal current(s) (where appropriate)		±
5.4.1.7	Input power		±
5.4.1.8	LO input power		±
5.4.1.9	Power dissipation		±
5.4.1.1	Power supply voltage(s)		+
5.4.1.2	Power supply current(s) (where appropriate)		+
5.4.1.3	Terminal voltage(s) (where appropriate)	+	+
5.4.1.4	Terminal current(s) (where appropriate)		+
5.4.1.5	Input power		+
5.4.1.6	LO input power		+
5.4.1.7	Power dissipation		+

The detail specification may indicate those values within table 2, including note 1 and note 2.

Parameters (see notes 1 and 2)	Symbols	Min.	Max.	Unit
NOTE 1 Where appropriate, in accordance with the type of considered circuit. NOTE 2 For power supply voltage range: limiting value(s) of the continuous voltage(s) at the supply terminal(s) with respect to a special electrical reference point; where appropriate, limiting value between specified supply terminals; when more than one voltage supply is required, a statement should be made as to whether the sequence in which these supplies are applied is significant: if so, the sequence should be stated; when more than one supply is needed, it may be necessary to state the combinations of ratings for these supply voltages and currents.				

5.4.2 Temperatures

- a) Operating temperature (ambient or reference-point temperature)
- b) Storage temperature
- c) Channel temperature
- d) Lead temperature (for soldering)

The detail specification may indicate those values within the table including the note.

Parameters (see note)	Symbols	Min.	Max.	Unit
NOTE Where appropriate, in accordance with the type of considered circuit.				

5.5 Operating conditions (within the specified operating temperature range)

These are not to be inspected, but may be used for quality assessment purposes.

5.5.1 Power supplies – Positive and/or negative values

5.5.2 Initialization sequences (where appropriate)

If special initialization sequences are necessary, the power supply sequencing and initialization procedure shall be specified.

5.5.3 Input voltage(s) or input signal (where appropriate)

5.5.4 Output current(s) (where appropriate)

5.5.5 Voltage and/or current of other terminal(s)

5.5.6 External elements (where appropriate)

5.5.7 Operating temperature range

5.6 Electrical characteristics

The characteristics shall apply over the full operating temperature range, unless otherwise specified.

Each characteristic ~~of 5.6~~ shall be stated: either

- a) over the specified range of operating temperatures, or
- b) at a temperature of 25 °C, and at maximum and minimum operating temperatures.

The parameters shall be specified corresponding to the type as shown in Table 3 below:

Table 3 – Electrical characteristics

Sub-clause	Parameters	Min.	Typical*	Max.	Types	
					A	B
5.6.1	Operating current		+	+	+	+
5.6.2	Conversion gain	+	+		+	+
5.6.3	Conversion gain flatness		+	+	+	+
5.6.4	LO/RF isolation	+	+		+	+
5.6.5	LO/IF isolation (where appropriate)	+	+		+	+
5.6.6	RF/IF isolation (where appropriate)	+	+		+	
5.6.7	IF/RF isolation (where appropriate)	+	+			+
5.6.8	Image rejection (where appropriate)	+	+		+	
5.6.9	Sideband suppression (where appropriate)	+	+			+
5.6.10	Output power at specified input power	+	+		+	+
5.6.11	Output power at 1 dB conversion compression	+	+		+	+
5.6.12	Noise figure		+	+	+	
5.6.13	Intermodulation distortion	±	+	+	+	+
5.6.14	Output power at the intercept point (for intermodulation products)	+	+		+	+
5.6.15	LO port return loss	+	+		+	+
5.6.16	RF port return loss	+	+		+	+
5.6.17	IF port return loss	+	+		+	+
* Optional.						
NOTE Conversion loss and conversion loss flatness can substitute for parameters 5.6.2 and 5.6.3.						

The detail specification may indicate these values within the table.

Characteristics	Symbols	Conditions	Min.	Typical*	Max.	Units
* Optional.						

5.7 Mechanical and environmental ratings, characteristics and data

Any specific mechanical and environmental ratings applicable shall be stated (see also 5.10 and 5.11 of IEC 60747-1, ~~Chapter VI, Clause 7~~).

5.8 Additional information

Where appropriate, the information detailed in the following subclauses shall be given.

5.8.1 Equivalent input and output circuit

Detail information shall be given regarding the type of input and output circuits, for example, input/output impedances, d.c. block, open-drain.

5.8.2 Internal protection

A statement shall be given to indicate whether the integrated circuit contains internal protection against high static voltages or electrical fields.

5.8.3 Capacitors at terminals

If capacitors for the input/output d.c. block are needed, these capacitances shall be stated.

5.8.4 Thermal resistance

5.8.5 Interconnections to other types of circuit

Where appropriate, details of the interconnections to other circuits shall be given.

5.8.6 Effects of externally connected component(s)

Curves or data indicating the effect of an externally connected component(s) that influence the characteristics may be given.

5.8.7 Recommendations for any associated device(s)

For example, decoupling of the power supply to a high frequency device shall be stated.

5.8.8 Handling precautions

Where appropriate, handling precautions specific to the circuit shall be stated (see also ~~IEC 60747-1, chapter IX~~ IEC 61340-5-1 and IEC 61340-5-2 concerning electrostatic-sensitive devices).

5.8.9 Application data

5.8.10 Other application information

5.8.11 Date of issue of the data sheet

6 Measuring methods

6.1 General

This clause prescribes measuring methods for electrical characteristics of integrated circuit microwave frequency converters.

6.1.1 General precautions

The general precautions listed in ~~Clause 4~~ 6.3, 6.4 and 6.6 of IEC 60747-1, ~~chapter VII~~, shall be applied. In addition, special care shall be taken to use low-ripple d.c. supplies and to decouple adequately all bias supply voltages at the frequency of measurement. Special care concerning the load impedance of the test circuit shall also be taken to measure the output power.

The input signal (RF for down-converter and IF for up-converter) level shall be a small signal condition under which the devices exhibit linear characteristics against the input signal, unless otherwise specified.

6.1.2 Characteristic impedance

The input and output characteristic impedances of the measurement systems, shown in the circuit in this standard, are 50 Ω . If they are not 50 Ω , they shall be specified.

6.1.3 Handling precautions

When handling electrostatic-sensitive devices, the handling precautions given in ~~IEC 60747-1, Chapter IX, Clause 4~~ IEC 61340-5-1 and IEC 61340-5-2, shall be observed.

6.1.4 Types

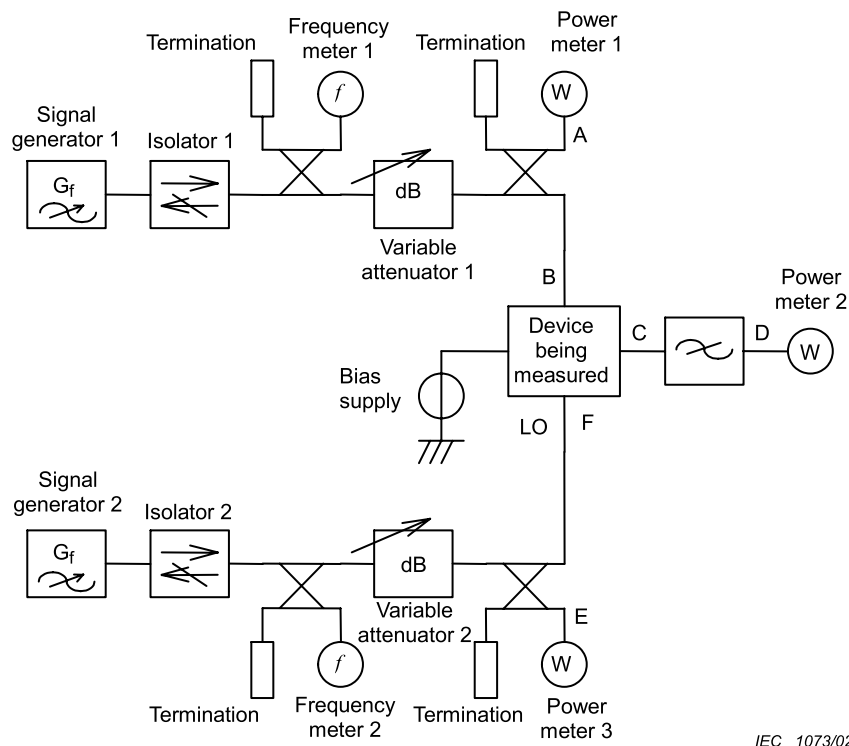
The devices in this standard are both package and chip types, measured using suitable test fixtures.

6.2 Conversion gain (G_c)

6.2.1 Purpose

To measure conversion gain under specified conditions.

6.2.2 Circuit diagram



IEC 1073/02

Figure 2 – Circuit diagram for the measurement of conversion gain

6.2.3 Principle of measurement

The conversion gain G_c , measured under the specified local oscillator power P_{LO} , is derived from the input power P_i and the output power P_o of the device, being measured as follows:

$$G_c = P_o - P_i \quad (1)$$

In the circuit diagram shown in Figure 2, P_i , P_o and P_{LO} are derived from the following equations:

$$P_i = P_1 - L_1 \quad (2)$$

$$P_o = P_2 + L_2 \quad (3)$$

$$P_{LO} = P_3 - L_3 \quad (4)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point C in dBm, minus the power at point D in dBm;

L_3 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1;

P_2 is the value indicated by power meter 2;

P_3 is the value indicated by power meter 3.

P_1 , P_0 , P_1 , P_2 , and P_3 are expressed in dBm. L_1 , L_2 , and L_3 are expressed in decibels (dB).

The conversion gain G_c is the power gain measured in the region where the change of output power in dBm is the same as that of the input power.

6.2.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatched at its input. The value of L_1 , L_2 , and L_3 shall be measured beforehand. The filter at the output rejects the undesired frequency band.

6.2.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during these measurements. Terminations shall be capable of handling the power in the test environment.

Harmonics or spurious responses of the signal generator shall be reduced to be negligible.

The conversion gain G_c shall be measured without the influence of filter impedance at LO and RF ports.

6.2.6 Measurement procedure

The frequency of the signal generator for the input signal shall be set to the specified value.

The frequency of the signal generator for the local signal shall be set to the specified value.

The power level of the local signal shall be adjusted to the specified value.

~~The bias conditions shall be applied as specified.~~ The bias under specified conditions is applied.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

The gain measured in this region is the conversion gain G_c .

6.2.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port
- Input frequency.

6.3 Conversion gain flatness (ΔG_c)

6.3.1 Purpose

To measure the conversion gain flatness under specified conditions.

6.3.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.3.3 Principle of measurement

See the principle of measurement of 6.2.3.

The linear gain flatness is derived from following equation

$$\Delta G_c = G_{c(\max)} - G_{c(\min)} \quad (5)$$

where $G_{c(\max)}$ and $G_{c(\min)}$ are the maximum and the minimum conversion gains in the specified frequency band at the specified input power, respectively.

There are two kinds of measurement methods of ΔG_c . One is the measurement under fixed local oscillator frequency and the other is the measurement under constant output frequency.

6.3.4 Circuit description and requirements

See the circuit description and requirements of 6.2.4.

6.3.5 Precaution to be observed

See the precaution to be observed of 6.2.5.

6.3.6 Measurement procedure

6.3.6.1 Conversion gain flatness for constant LO frequency

The frequency of the signal generator for the local signal shall be set to the specified value.

The frequency of the signal generator for the input signal shall be set to the specified value.

~~The bias conditions shall be applied as specified.~~ The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

~~Decide on~~ Set the suitable input power level for measuring the conversion gain.

Vary the input frequency in the specified frequency band while keeping the input power level constant.

Obtain the maximum conversion gain $G_{c(\max)}$ and the minimum conversion gain $G_{c(\min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.6.2 Conversion gain flatness for constant output frequency

The frequency of the signal generator for the local signal shall be set to obtain the specified output frequency.

The frequency of the signal generator for the input signal within the specified input frequency range shall be set to obtain the specified output frequency.

~~The bias conditions shall be applied as specified.~~ The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

~~Decide on~~ Set the suitable input power level for measuring the conversion gain.

Vary the input and local frequencies within the specified frequency band ~~at a constant input power level.~~ while keeping the input power level and the local power level constant.

Obtain the maximum conversion gain $G_{c(max)}$ and the minimum conversion gain $G_{c(min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.6.3 Conversion gain flatness for constant input frequency

The frequency of the signal generator for the local signal shall be set to the specified value.

The frequency of the signal generator for the input signal shall be set to the specified value.

The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

Set the suitable input power level for measuring the conversion gain.

Vary the local frequency within the specified frequency band while keeping the local power level constant.

Obtain the maximum conversion gain $G_{c(max)}$ and the minimum conversion gain $G_{c(min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.7 Specified conditions

6.3.7.1 Conversion gain flatness for constant LO frequency

– Ambient or reference point temperature

- Bias conditions
- Frequency of local signal
- Incident power of local port
- Input frequency band.

6.3.7.2 Conversion gain flatness for constant output frequency

- Ambient or reference point temperature
- Bias conditions
- Frequency band of local signal
- Incident power of local port
- ~~Input power~~
- Input frequency range
- Output frequency.

6.3.7.3 Conversion gain flatness for constant input frequency

- Ambient or reference point temperature
- Bias conditions
- Frequency band of local signal
- Incident power of local port
- Input frequency

6.4 LO/IF isolation ($P_{LO}/P_{LO(IF)}$)

6.4.1 Purpose

To measure the isolation from the local port to the IF port under specified conditions.

6.4.2 Circuit diagram

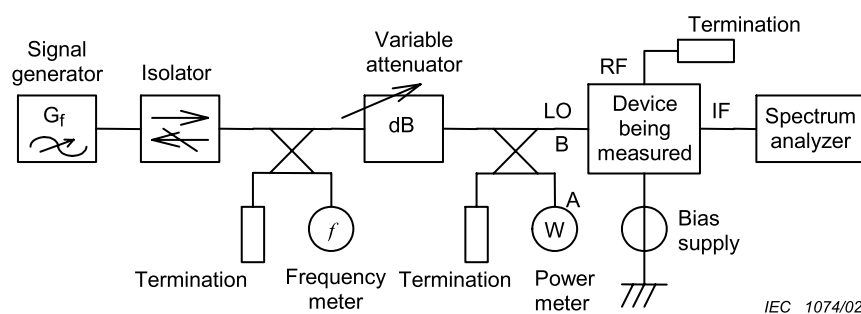


Figure 3 – Circuit diagram for the measurement of the LO/IF isolation

6.4.3 Principle of measurement

In the circuit diagram shown in Figure 3, the input local power P_{LO} in dBm is derived as follows:

$$P_{LO} = P_1 - L_1 \quad (6)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

P_1 is the value indicated by the power meter in dBm.

$P_{LO(IF)}$ is the local frequency component of the output power at the IF port in dBm when the specified P_{LO} in dBm is applied.

The local port to IF port isolation $P_{LO}/P_{LO(IF)}$ in decibels (dB) is derived as follows:

$$P_{LO}/P_{LO(IF)} = P_{LO} - P_{LO(IF)} \quad (7)$$

where $P_{LO(IF)}$ is the value indicated by the spectrum analyzer.

6.4.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its local ports.

L_1 shall be measured beforehand.

NOTE The spectrum analyzer can be replaced with a power meter and a filter having proper frequency response and impedance characteristics. In the case of such a set-up, the filter must reject the undesired frequency components, and should not change the terminal impedances for frequencies such as RF, LO and IF.

6.4.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during the measurement.

Harmonics or spurious responses of the signal generator shall be reduced so as to be negligible.

6.4.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} , which is given by equation (6), is applied.

The power level of the local frequency component at the IF port $P_{LO(IF)}$ is measured by the spectrum analyzer.

The local port to IF port isolation $P_{LO}/P_{LO(IF)}$ is calculated from equation (7).

6.4.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port.

6.5 LO/RF isolation ($P_{LO}/P_{LO(RF)}$)

6.5.1 Purpose

To measure the isolation from the local port to the RF port under specified conditions.

6.5.2 Circuit diagram

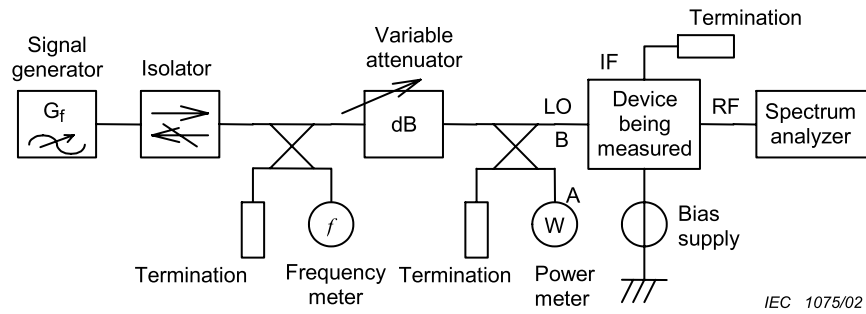


Figure 4 – Circuit diagram for the measurement of the LO/RF isolation

6.5.3 Principle of measurement

In the circuit diagram shown in Figure 4, the input local power P_{LO} in dBm is derived as follows:

$$P_{LO} = P_1 - L_1 \quad (8)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

P_1 is the value indicated by the power meter in dBm.

$P_{LO(RF)}$ is the local frequency component of the output power at the RF port in dBm when the specified P_{LO} in dBm is applied.

The local port to RF port isolation $P_{LO}/P_{LO(RF)}$ in decibels (dB) is derived as follows:

$$P_{LO}/P_{LO(RF)} = P_{LO} - P_{LO(RF)} \quad (9)$$

where $P_{LO(RF)}$ is the value indicated by the spectrum analyzer.

6.5.4 Circuit description and requirements

See the circuit description and requirements described in 6.4.4.

6.5.5 Precautions to be observed

See the precautions to be observed described in 6.4.5.

6.5.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power, P_{LO} , which is given by equation (8), is applied.

The power level of the local frequency component at the RF port $P_{LO(RF)}$ is measured by the spectrum analyzer.

The local port to RF port isolation $P_{LO}/P_{LO(RF)}$ is calculated from equation (9).

6.5.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port
- Termination condition at IF port.

6.6 RF/IF, IF/RF isolation

6.6.1 RF/IF isolation ($P_{RF}/P_{RF(IF)}$) – Type A

6.6.1.1 Purpose

To measure the isolation from the RF port to the IF port under specified conditions.

6.6.1.2 Circuit diagram

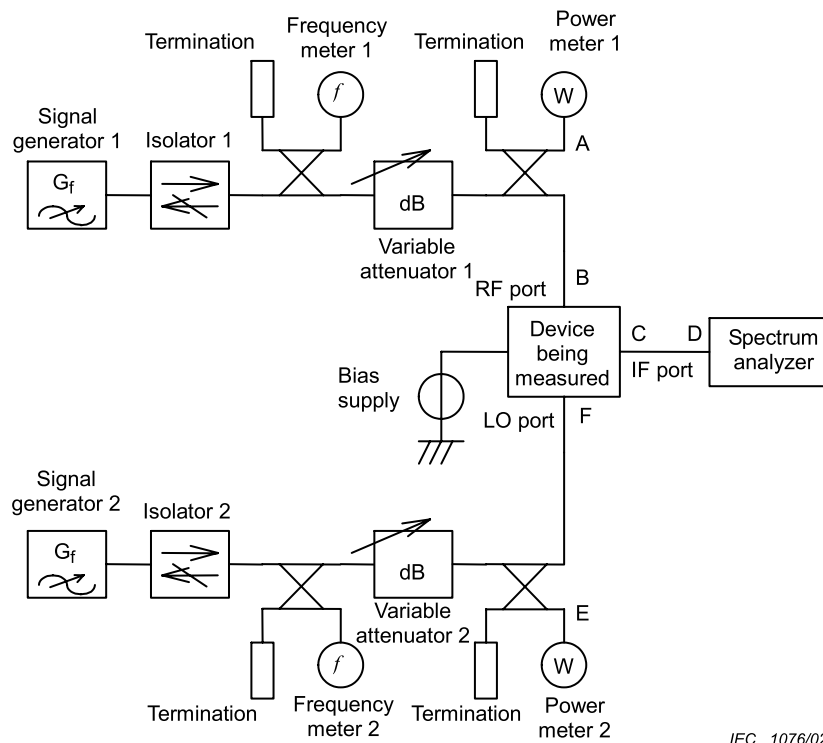


Figure 5 – Circuit diagram for the measurement of the RF/IF isolation for type A

6.6.1.3 Principle of measurement

In the circuit diagram shown in Figure 5, the input RF power P_{RF} and the LO power P_{LO} are derived as follows:

$$P_{RF} = P_1 - L_1 \quad (10)$$

$$P_{LO} = P_2 - L_2 \quad (11)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 at the RF port in dBm;

P_2 is the value indicated by power meter 2 at the LO port in dBm.

$P_{\text{RF(IF)}}$ is the RF frequency component of the output power at the IF port in dBm when the specified P_{RF} and P_{LO} are applied.

The isolation $P_{\text{RF}}/P_{\text{RF(IF)}}$ from the RF port to the IF port is derived as follows:

$$P_{\text{RF}}/P_{\text{RF(IF)}} = P_{\text{RF}} - P_{\text{RF(IF)}} \quad (12)$$

where $P_{\text{RF(IF)}}$ is the value indicated by the spectrum analyzer in dBm.

6.6.1.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its RF and LO ports.

L_1 and L_2 shall be measured beforehand.

6.6.1.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during measurement.

Harmonics or spurious responses of the signal generators shall be reduced so as to be negligible.

6.6.1.6 Measurement procedure

The frequency of the signal generator 1 for the RF input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The local power P_{LO} , which is given by equation (11), is applied and adjusted to the specified value by the variable attenuator 2.

The RF power P_{RF} , which is given by equation (10), is applied and adjusted to the specified value by the variable attenuator 1.

The power level of the RF signal component at the IF port $P_{\text{RF(IF)}}$ is measured by the spectrum analyzer.

The isolation $P_{\text{RF}}/P_{\text{RF(IF)}}$ from the RF port to the IF port is calculated from the equation (12).

6.6.1.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Input power of RF port
- Frequency of RF signal.

6.6.2 ~~RF/IF~~ IF/RF isolation ($P_{IF}/P_{IF(RF)}$) – Type B

6.6.2.1 Purpose

To measure the isolation from the IF port to the RF port under specified conditions.

6.6.2.2 Circuit diagram

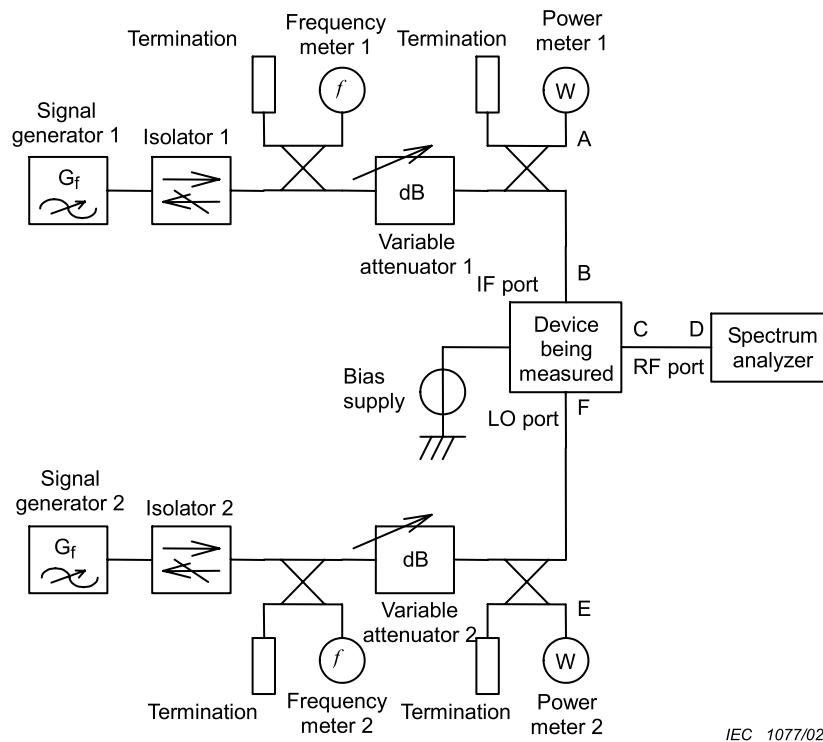


Figure 6 – Circuit diagram for the measurement of the ~~RF/IF~~ IF/RF isolation for type B

6.6.2.3 Principle of measurement

In the circuit diagram shown in Figure 6, the input IF power P_{IF} and the LO power P_{LO} are derived as follows:

$$P_{IF} = P_1 - L_1 \quad (13)$$

$$P_{LO} = P_2 - L_2 \quad (14)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 at the IF port in dBm;

P_2 is the value indicated by power meter 2 at the LO port in dBm.

$P_{\text{IF(RF)}}$ is the IF frequency component of the output power at the RF port in dBm when the specified P_{IF} and P_{LO} are applied.

The isolation $P_{\text{IF}}/P_{\text{IF(RF)}}$ from the IF port to the RF port is derived as follows:

$$P_{\text{IF}}/P_{\text{IF(RF)}} = P_{\text{IF}} - P_{\text{IF(RF)}} \quad (15)$$

where $P_{\text{IF(RF)}}$ is the value indicated by the spectrum analyzer in dBm.

6.6.2.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its IF and LO port.

L_1 and L_2 shall be measured beforehand.

6.6.2.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during measurement.

Harmonics or spurious responses of the signal generators shall be reduced to be negligible.

6.6.2.6 Measurement procedure

The frequency of the signal generator 1 for the IF input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The local power P_{LO} , which is given by equation (14), is applied and adjusted to the specified value by the variable attenuator 2.

The IF power P_{IF} , which is given by equation (13), is applied and adjusted to the specified value by the variable attenuator 1.

The power level of the IF signal component at the RF port $P_{\text{IF(RF)}}$ is measured by the spectrum analyzer.

The isolation $P_{\text{IF}}/P_{\text{IF(RF)}}$ from the IF port to the RF port is calculated from equation (15).

6.6.2.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Input power of IF port
- Frequency of IF signal.

6.7 Image rejection ($P_o / P_{o(im)}$)**6.7.1 Purpose**

To measure the image rejection under specified conditions.

6.7.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.7.3 Principle of measurement

The image frequency $f_{(image)}$ is defined as follows:

$$f_{(image)} = 2f_{(local)} - f_{(rf)} \quad (16)$$

where

$f_{(local)}$ is the specified local frequency;

$f_{(rf)}$ is the specified RF frequency.

The input power P_i and the local power P_{LO} are as follows:

$$P_i = P_1 - L_1 \quad (17)$$

$$P_{LO} = P_3 - L_3 \quad (18)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_3 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 in dBm;

P_3 is the value indicated by power meter 3 in dBm.

P_i , P_{LO} , P_1 and P_3 are expressed in dBm. L_1 and L_3 are expressed in decibels (dB).

The image rejection $P_o / P_{o(im)}$ in decibels (dB) is derived as follows:

$$P_o / P_{o(im)} = P_{o(rf)} - P_{o(image)} \quad (19)$$

where

$P_{o(rf)}$ is the value indicated by power meter 2 in dBm when the input frequency and power are the specified RF frequency and input power;

$P_{o(image)}$ is the value indicated by power meter 2 in dBm when the input frequency and power are the image frequency and input power.

6.7.4 Circuit description and requirements

See the circuit description and requirements of 6.2.4.

6.7.5 Precaution to be observed

See the precaution to be observed of 6.2.5.

6.7.6 Measurements procedure

The frequency of the signal generator for the input signal shall be adjusted to the specified RF frequency.

The power level of the input signal (P_i) shall be adjusted to the specified input power.

The frequency of the signal generator for the local signal shall be adjusted to the specified local frequency.

The power level of the local signal (P_{LO}) shall be adjusted to the specified local power.

The bias under specified conditions is applied to the device being measured.

The output power ($P_{o(rf)}$) is measured by power meter 2.

The frequency of the signal generator for the input signal shall be changed to the specified image frequency.

The power level of the input signal (P_i) shall be adjusted to the specified input power.

The output power ($P_{o(image)}$) is measured by power meter 2.

The image rejection is calculated from equation (19).

6.7.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- RF frequency
- Local frequency
- Input power of RF port
- Local power.

6.8 Sideband suppression ($P_o / P_{o(U)}$)

6.8.1 Purpose

To measure the sideband suppression under specified conditions.

6.8.2 Circuit diagram

See the circuit diagram shown in Figure 6.

6.8.3 Principle of measurement

The incident power of the IF port P_{IF} and the local power P_{LO} are derived as follows:

$$P_{IF} = P_1 - L_1 \quad (20)$$

$$P_{LO} = P_2 - L_2 \quad (21)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 in dBm;

P_2 is the value indicated by power meter 2 in dBm.

The sideband suppression $P_o/P_{o(U)}$ is derived as follows:

$$P_o/P_{o(U)} = P_o - P_{o(U)} \quad (22)$$

In the circuit diagram shown in Figure 6, the output power of the desired sideband P_o and the output power of the undesired sideband $P_{o(U)}$ are derived from the following equations:

$$P_o = P_3 + L_3 \quad (23)$$

$$P_{o(U)} = P_4 + L_4 \quad (24)$$

where

L_3 is the power at point C for the desired sideband frequency in dBm, minus the power at point D for the desired sideband frequency in dBm;

L_4 is the power at point C for the undesired sideband frequency in dBm, minus the power at point D for the undesired sideband frequency in dBm;

P_3 is the value for the desired sideband frequency indicated by the spectrum analyzer in dBm;

P_4 is the value for the undesired sideband frequency indicated by the spectrum analyzer in dBm.

6.8.4 Circuit description and requirements

See the circuit description and requirements of 6.6.2.4. The circuit losses L_1 , L_2 , L_3 and L_4 shall be measured beforehand.

6.8.5 Precautions to be observed

See the precautions to be observed in 6.6.2.5.

6.8.6 Measurement procedure

The IF frequency of the signal generator 1 shall be adjusted to the specified value.

The local frequency of the signal generator 2 shall be adjusted to the specified value.

The bias under specified conditions is applied to the device being measured.

The incident power of the IF port P_{IF} shall be adjusted to the specified value.

The local power P_{LO} shall be adjusted to the specified value.

The value of P_3 is measured with the spectrum analyzer.

The value of P_4 is measured with the spectrum analyzer.

The sideband suppression $P_o/P_{o(U)}$ is calculated from equations (22), (23) and (24).

6.8.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Incident power of IF port
- IF frequency
- Local power
- Local frequency.

6.9 Output power (P_o)

6.9.1 Purpose

To measure the output power under specified conditions.

6.9.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.9.3 Principle of measurement

In the circuit diagram shown in Figure 1, the output power P_o under the specified local signal and the input signal is derived from equation (3).

6.9.4 Circuit description and requirements

See the circuit description and requirements described in 6.2.4.

6.9.5 Precautions to be observed

See the precautions to be observed described in 6.2.5.

6.9.6 Measurement procedure

The frequency of the signal generator 1 for the input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} is applied.

The specified input power P_i is applied.

The power level P_2 is measured.

The output power is calculated from equation (3).

6.9.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of input signal
- Power level of input signal
- Local power
- Frequency of local signal.

6.10 Output power at 1-dB conversion compression ($P_{o(1dB)}$)

6.10.1 Purpose

To measure the output power at 1-dB conversion compression under specified conditions.

6.10.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.10.3 Principle of measurement

See the principle of measurement in 6.2.3.

The output power at 1-dB conversion compression $P_{o(1dB)}$ is the output power where the conversion gain decreases by 1 dB compared with G_c .

6.10.4 Circuit description and requirements

See the circuit description and requirements described in 6.2.4.

6.10.5 Precautions to be observed

See the precautions to be observed described in 6.2.5.

6.10.6 Measurement procedure

The frequency of the signal generator 1 for the input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} is applied.

The adequate input power is applied.

By varying the input power, confirm the change of conversion gain in decibels is the same as that of input power.

The input power is increased till the conversion gain decreases by 1 dB, compared with G_c .

The power level P_2 is measured.

The output power at 1-dB conversion compression $P_{o(1dB)}$ is calculated from equation (3).

6.10.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of input signal
- Local power
- Frequency of local signal.

6.11 Noise figure (F)

6.11.1 Purpose

To measure the noise figure under specified conditions.

6.11.2 Circuit diagram

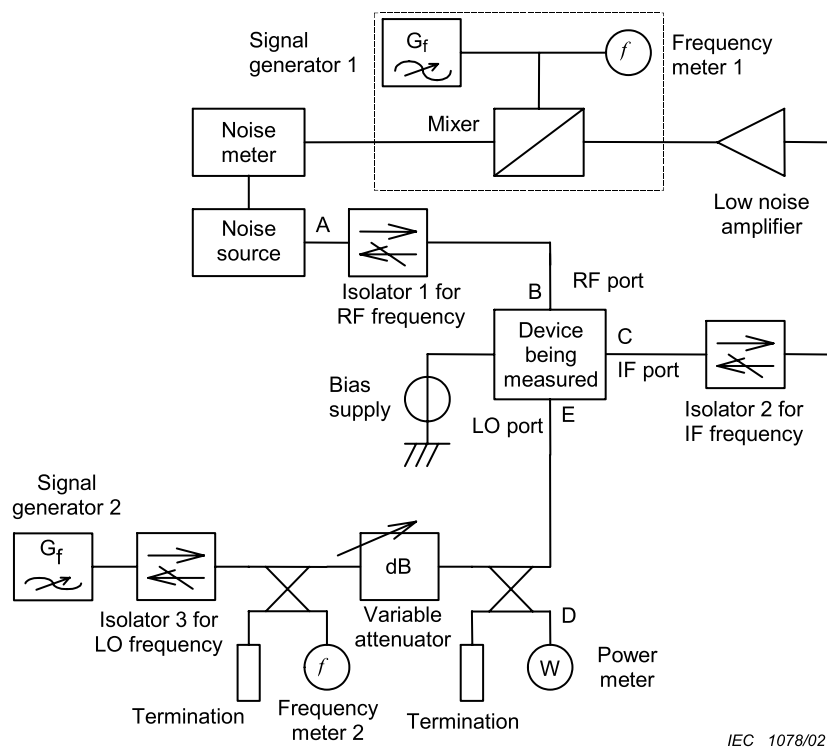


Figure 7 – Circuit diagram for measurement of noise figure

6.11.3 Principle of measurement

The noise figure F of the device being measured is derived from the following equation, when the oscillation power P_{LO} is applied to the device being measured:

$$F = 10 \log \left(10^{(F_{12} - L_1) / 10} - \frac{10^{(F_2 / 10)} - 1}{10^{(G_c / 10)}} \right) \quad (25)$$

The noise figure measurement is carried out by using the hot and cold measurement method; F_{12} , F_2 , and G_c are calculated as follows:

$$F_{12} = 10 \log \left(\frac{10^{(ENR_1 / 10)}}{(P_{N1}/P_{N2}) - 1} \right) \quad (26)$$

$$F_2 = 10 \log \left(\frac{10^{(ENR_2 / 10)}}{(P_{N3}/P_{N4}) - 1} \right) \quad (27)$$

$$G_c = 10 \log \left(\frac{P_{N1} - P_{N2}}{P_{N3} - P_{N4}} \right) + (ENR_1 - ENR_2) + L_1 \quad (28)$$

The input local power P_{LO} is derived as follows:

$$P_{LO} = P_1 - L_2 \quad (29)$$

where

- F_{12} is the overall noise figure in decibels (dB);
- F_2 is the noise figure in decibels (dB) after point C;
- ENR_1 is the excess noise ratio of the noise source at the RF frequency;
- ENR_2 is the excess noise ratio of the noise source at the IF frequency;
- P_{N1} is the measured noise powers in watts (W) under the hot state of the noise source;
- P_{N2} is the measured noise powers in watts (W) under the cold state of the noise source;
- P_{N3} is the measured noise powers in watts (W) under the hot state of the noise source in the case of directly connecting point A to C;
- P_{N4} is the measured noise powers in watts (W) under the cold state of the noise source in the case of directly connecting point A to C;
- P_{LO} is the value indicated by the power meter at the LO port in decibels (dB);
- L_1 is the circuit loss in decibels (dB) between point A and B;
- L_2 is the power at point D in decibels (dB), minus the power at point E in decibels (dB).

The temperature of the measurement is 290 K.

6.11.4 Circuit description and requirements

The purpose of isolator 1 and isolator 3 is to enable the power level to the device being measured to be kept constant, irrespective of impedance mismatches at its RF and LO ports.

The purpose of isolator 2 is to enable the power level to the low noise amplifier to be kept constant, irrespective of impedance mismatches.

When the IF frequency is within the noise meter operating range, the circuit surrounded by the dotted line can be omitted.

When the RF frequency and the IF frequency are not covered by the one noise source frequency range, the noise source for the IF frequency and the noise source for the RF frequency are needed.

The circuit loss L_1 and L_2 shall be measured beforehand.

6.11.5 Precautions to be observed

The entire circuit shall be shielded and earthed to prevent undesired signals. For noise figure measurement under the SSB condition, careful attention shall be paid to the image and to other spurious responses that are generated by the mixer and device being measured.

These spurious responses shall be reduced so as to be negligible.

6.11.6 Measurement procedure

The frequency of the signal generator 1 is adjusted to meet the specified condition.

NOTE The measurement procedure is omitted when the mixer circuit is not used.

In order to measure the noise contribution of the measurement system, connect point A to C in Figure 7 without the device being measured.

The noise powers P_{N3} and P_{N4} corresponding to the hot and cold noise source, respectively, are measured.

The device being measured is inserted as shown in Figure 7.

The frequency of the signal generator 2 is adjusted to meet the specified condition.

The bias under specified conditions is applied.

The P_{LO} under specified conditions are applied.

The noise powers P_{N1} and P_{N2} corresponding to the hot and cold noise source, respectively, are measured.

The noise figure in decibels (dB) is calculated by equation (25).

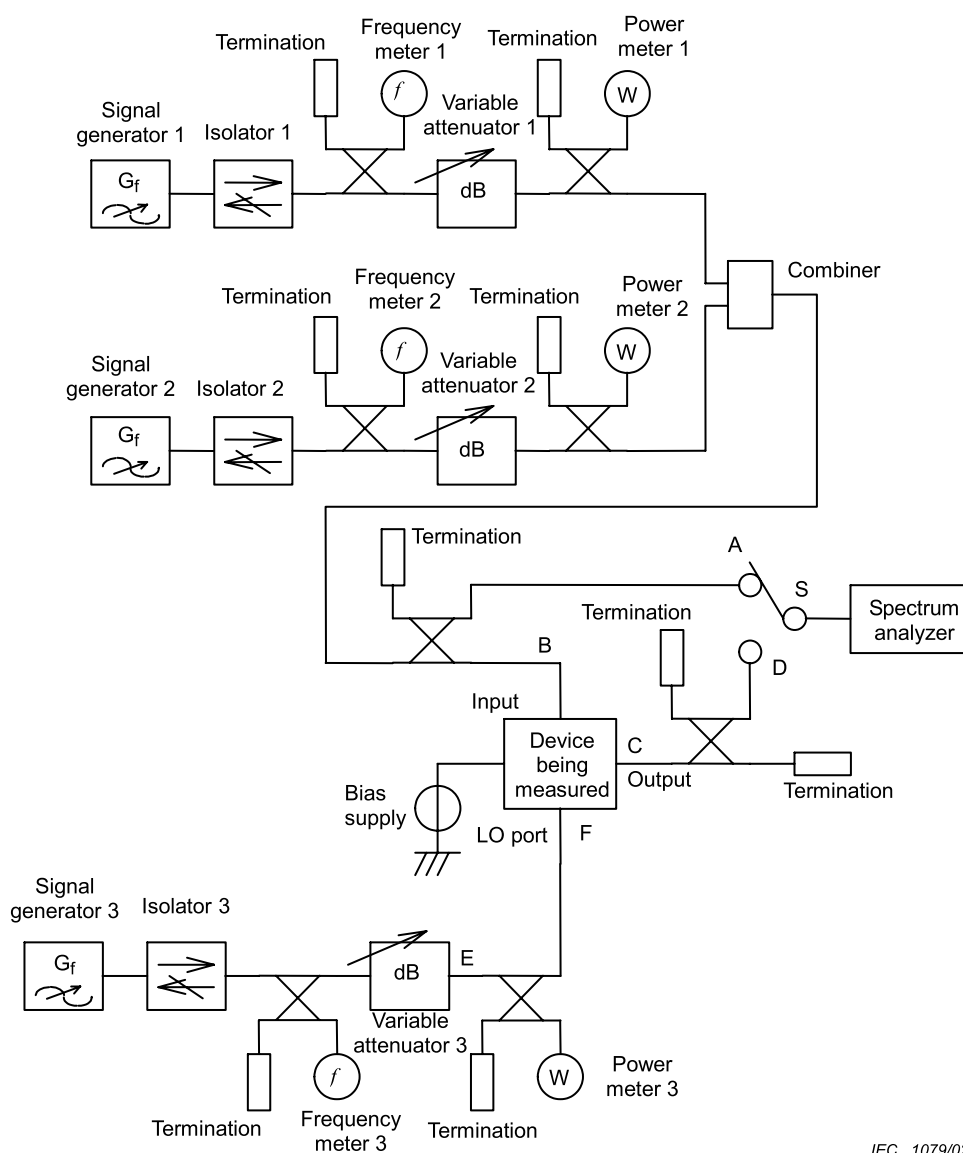
6.11.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Single-side band or double-side band.

6.12 Intermodulation distortion (P_n/P_1)

6.12.1 Purpose

To measure the intermodulation distortion under specified conditions.



$$P_{\text{LO}} = P_4 - L_3 \quad (33)$$

P_1 , P_2 and P_3 are the values indicated by the spectrum analyzer and correspond to P_i , P_o and P_n , respectively. P_4 is the output power at point E calculated from the value indicated by signal generator 3.

L_1 , L_2 and L_3 are the circuit losses from point A to point B, from point C to point D and from point E to point F shown in Figure 8, respectively.

P_i , P_o , P_n and P_{LO} are expressed in dBm. ~~P_4/P_n~~ L_1 , L_2 and L_3 are expressed in decibels (dB).

The intermodulation distortion (P_n/P_1) in ~~decibels~~ dBc is derived from equations (31) and (32) as follows:

$$P_n/P_1 = P_n - P_o = P_3 - P_2 \quad (34)$$

6.12.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its ports.

L_1 shall be measured beforehand.

The input and output ports correspond to the RF and IF ports for type A devices, and to the IF and RF ports for type B devices, respectively.

6.12.5 Precautions to be observed

See the precautions to be observed in 6.4.5.

It is better to terminate port D when the switch is connected to position A, and vice versa.

6.12.6 Measurement procedure

The bias under specified conditions is applied.

The signal generator 3 is tuned on and the LO signal is added to the device being measured with the specified level P_4 using the variable attenuator 3.

The switch is connected to position A.

The signal generator 1 is tuned on and the input signal is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 1.

The signal generator 2 is tuned on and the input signal is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 2.

The switch is connected to position D.

The output power P_2 and the intermodulation products P_3 in dBm are measured using the spectrum analyzer.

The intermodulation distortion on the specified input power P_1 is derived from equations (30) to (34).

6.12.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Input power
- Local power
- Input frequency
- Local frequency
- n th order.

6.13 Output power at the intercept point (for intermodulation products) ($P_{n(IP)}$)**6.13.1 Purpose**

To measure the output power at the intercept point (for intermodulation products) under specified conditions.

6.13.2 Circuit diagram

See the circuit diagram of 6.12.2.

6.13.3 Principle of measurement

Refer to the principle of measurement of 6.12.3.

6.13.4 Circuit description and requirements

See the circuit description and requirements of 6.12.4.

6.13.5 Precautions to be observed

See the precautions to be observed in 6.12.5.

6.13.6 Measurement procedure

The bias under specified conditions is applied.

The signal generator 3 is tuned on and the LO signal is added to the device being measured with the specified level P_4 using the variable attenuator 3.

The switch is connected to position A.

The signal generator 1 is tuned on and the input power is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 1.

The signal generator 2 is tuned on and the input power is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 2.

The switch is connected to position D.

The output power P_2 and the intermodulation products P_3 in dBm are measured using the spectrum analyzer.

Changing the power level of the input power using the variable attenuators 1 and 2, the above procedure is repeated within the specified range.

The obtained data are plotted on linear scales.

The straight lines of the output power and intermodulation products in the linear region are extended (slope of 1 and 3 are respectively observed).

The output power where the two extended lines intercept is the output power at the intercept point (for intermodulation products) under specified conditions.

6.13.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Input power range
- Local power
- Input frequency
- Local frequency
- n th order.

6.14 LO port return loss ($L_{\text{ret(LO)}}$)

6.14.1 Purpose

To measure the LO port return loss under specified conditions.

6.14.2 Circuit diagram

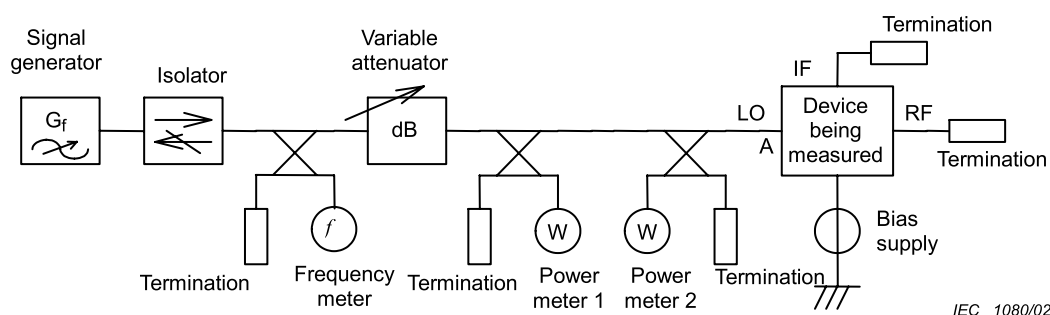


Figure 9 – Circuit for the measurement of the LO port return loss

6.14.3 Principle of measurement

In the circuit diagram shown in Figure 9, the LO port return loss is derived from the following equation:

$$L_{\text{ret(LO)}} = P_1 - P_2 \quad (35)$$

where

- P_1 is the value indicated by power meter 2 in dBm, when the incident LO power is totally reflected at point A;
- P_2 is the value indicated by power meter 2 in dBm, when the LO port of the device being measured is connected to point A.

6.14.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant, irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and other ports are connected to the termination.

The directivity of the directional coupler is enough compared to the return loss of the device being measured.

6.14.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during these measurements. The terminations shall be capable of handling the power in the test environment.

Harmonics or spurious responses of the signal generator shall be reduced to be negligible.

6.14.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

Point A is shunted.

Then, the power of power meter 2 is measured. This power is P_1 .

The LO port of the device being measured is then connected with point A and the other ports are connected to the termination.

The bias under specified conditions is applied.

Then, the power of power meter 2 is measured. This power is P_2 .

The LO port return loss is derived from equation (35).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.14.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of LO signal
- Incident power at the LO port.

6.15 RF port return loss ($L_{\text{ret(RF)}}$)

6.15.1 Purpose

To measure the RF port return loss under specified conditions.

6.15.2 Circuit diagram

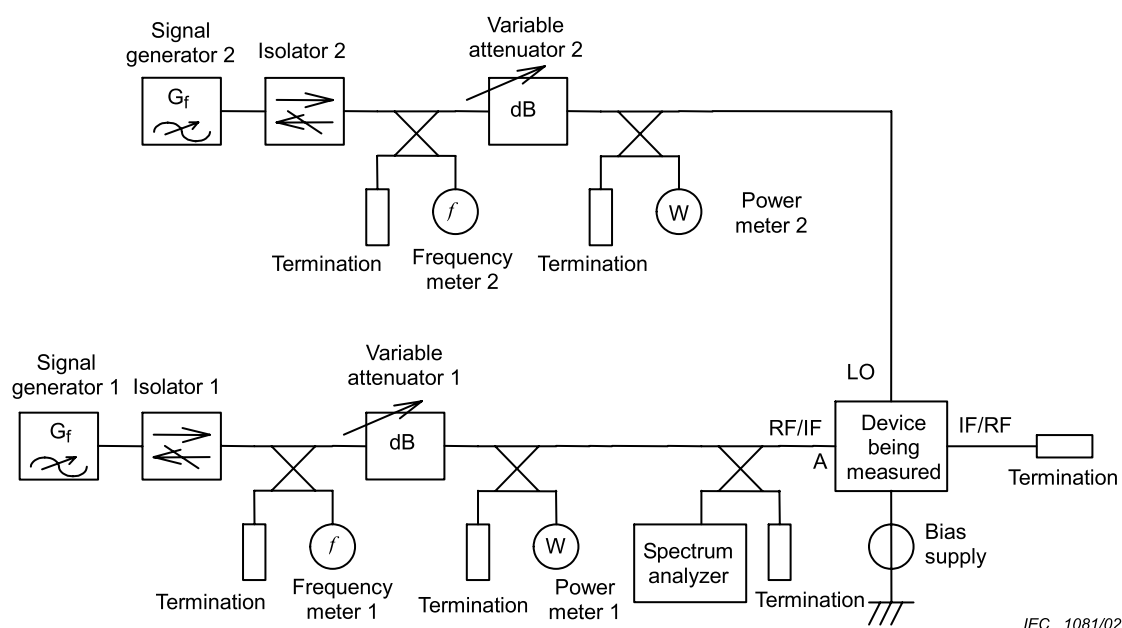


Figure 10 – Circuit for the measurement of the RF/IF port return loss

6.15.3 Principle of measurement

In the circuit diagram shown in Figure 10, the RF port return loss is derived from the following equation:

$$L_{\text{ret(RF)}} = P_1 - P_2 \quad (36)$$

where

P_1 is the value indicated by the spectrum analyzer in dBm, when the incident RF power is totally reflected at point A;

P_2 is the value indicated by the spectrum analyzer in dBm, when the RF port of the device being measured is connected to point A.

6.15.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and the other ports are connected to the terminations.

The directivity of the directional coupler is sufficient compared to the return loss of the device being measured.

6.15.5 Precautions to be observed

See the precautions to be observed in 6.2.5.

6.15.6 Measurement procedure

The frequency of the signal generator 1 for the RF signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the LO signal shall be adjusted to the specified value.

The power level of the LO signal shall be adjusted to the specified value.

Point A is shunted.

Then, the power of the spectrum analyzer is measured. This power is P_1 .

The RF port of the device being measured is then connected with point A.

The bias under specified conditions is applied.

Then, the power of the spectrum analyzer is measured. This power is P_2 .

The RF port return loss is derived from equation (36).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.15.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of RF signal
- Incident power at the RF port
- Frequency of LO signal
- Incident power at the LO port.

6.16 IF port return loss ($L_{\text{ret(IF)}}$)**6.16.1 Purpose**

To measure the IF port return loss under specified conditions.

6.16.2 Circuit diagram

See the circuit diagram shown in Figure 10.

6.16.3 Principle of measurement

In the circuit diagram shown in Figure 10, the IF port return loss is derived from following equation:

$$L_{\text{ret(IF)}} = P_1 - P_2 \quad (37)$$

where

P_1 is the value indicated by the spectrum analyzer in dBm, when the incident IF power is totally reflected at point A;

P_2 is the value indicated by the spectrum analyzer in dBm, when the IF port of the device being measured is connected to point A.

6.16.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and other ports are connected to the terminations.

The directivity of the directional coupler is sufficient compared to the return loss of the device being measured.

6.16.5 Precautions to be observed

See the precautions to be observed in 6.2.5.

6.16.6 Measurement procedure

The frequency of the signal generator 1 for the IF signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the LO signal shall be adjusted to the specified value.

The power level of the LO signal shall be adjusted to the specified value.

Point A is shunted.

The power of the spectrum analyzer is then measured. This power is P_1 .

Then, the IF port of the device being measured is connected with point A.

The bias under specified conditions is applied.

The power of the spectrum analyzer is then measured. This power is P_2 .

The IF port return loss is derived from equation (37).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.16.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of IF signal
- Incident power at the IF port
- Frequency of LO signal
- Incident power at the LO port.

Bibliography

IEC 60747-4:2007, *Semiconductor devices – Discrete devices – Part 4: Microwave diodes and transistors*
IEC 60747-4:2007/AMD 1:2017

FINAL VERSION

**Semiconductor devices –
Part 16-3: Microwave integrated circuits – Frequency converters**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

SEMICONDUCTOR DEVICES –

**Part 16-3: Microwave integrated circuits –
Frequency converters**

FOREWORD

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This Consolidated version of IEC 60747-16-3 bears the edition number 1.2. It consists of the first edition (2002-05) [documents 47E/212/FDIS and 47E/219/RVD], its amendment 1 (2009-03) [documents 47E/357/CDV and 47E/372/RVC] and its amendment 2 (2017-08) [documents 47E/545/CDV and 47E/562/RVC]. The technical content is identical to the base edition and its amendments.

This Final version does not show where the technical content is modified by amendments 1 and 2. A separate Redline version with all changes highlighted is available in this publication.

International Standard IEC 60747-16-3 has been prepared by subcommittee 47E: Discrete semiconductor devices, of IEC technical committee 47: Semiconductor devices.

The committee has decided that the contents of the base publication and its amendments will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

SEMICONDUCTOR DEVICES –**Part 16-3: Microwave integrated circuits –
Frequency converters****1 Scope**

This part of IEC 60747 provides new measuring methods, terminology and letter symbols, as well as essential ratings and characteristics for integrated circuit microwave frequency converters.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-702, *International Electrotechnical Vocabulary – Chapter 702: Oscillations, signals and related devices* (available at < <http://www.electropedia.org/> >)

IEC 60617, *Graphical symbols for diagrams* (available at < <http://std.iec.ch/iec60617> >)

IEC 60747-1:2006, *Semiconductor devices – Part 1: General*
IEC 60747-1:2006/AMD 1:2010

IEC 60748-2:1997, *Semiconductor devices – Integrated circuits – Part 2: Digital integrated circuits*

IEC 60748-3, *Semiconductor devices – Integrated circuits – Part 3: Analogue integrated circuits*

IEC 60748-4, *Semiconductor devices – Integrated circuits – Part 4: Interface integrated circuits*

IEC 61340-5-1:2007, *Electrostatics – Part 5-1: Protection of electronic devices from electrostatic phenomena – General requirements*

IEC/TR 61340-5-2:2007, *Electrostatics – Part 5-2: Protection of electronic devices from electrostatic phenomena – User guide*

3 Terms and definitions

For the purpose of this part of IEC 60747, the following terms and definitions apply:

3.1**conversion gain, G_c**

ratio of the desired converted output power to the input power

NOTE Usually, the conversion gain is expressed in decibels.

3.2**conversion gain flatness, ΔG_c**

difference between the maximum and the minimum conversion gain for a specified input power in a specified frequency range

3.3

LO/RF isolation, $P_{LO}/P_{LO(RF)}$

ratio of the incident local power to the local leakage power at the RF port with the IF port terminated in a specified impedance

3.4

LO/IF isolation, $P_{LO}/P_{LO(IF)}$

ratio of the incident local power to the local leakage power at the IF port with the RF port terminated in a specified impedance

3.5

RF/IF isolation, $P_{RF}/P_{RF(IF)}$

ratio of the incident RF power to the RF feedthrough power at the IF port for a specified local power

NOTE Usually, the RF/IF isolation is applied to the down-converter.

3.6

IF/RF isolation, $P_{IF}/P_{IF(RF)}$

ratio of the incident IF power to the IF feedthrough power at the RF port for a specified local power

NOTE Usually, the IF/RF isolation is applied to the up-converter.

3.7

image rejection, $P_o/P_{o(im)}$

ratio of the output power when the RF signal is applied, to the output power when the image signal is applied

NOTE Usually, the image rejection is applied to the down-converter.

3.8

sideband suppression, $P_o/P_{o(U)}$

ratio of the output power of the desired sideband to the output power of the undesired sideband

NOTE Usually, the sideband suppression is applied to the up-converter.

3.9

LO port return loss, $L_{ret(LO)}$

ratio of the specified incident power at the LO port to the reflected power at the LO port, with the RF port and the IF port terminated in each specified impedance

3.10

RF port return loss, $L_{ret(RF)}$

ratio of the incident power at the RF port to the reflected power at the RF port for a specified local power, with the IF port terminated in a specified impedance

3.11

IF port return loss, $L_{ret(IF)}$

ratio of the incident power at the IF port to the reflected power at the IF port for a specified local power, with the RF port terminated in a specified impedance

3.12

output power

P_o

RF power measured at the output port

[SOURCE: IEC 60747-4:2007/AMD1:2017, 7.2.2]

3.13**output power at 1-dB conversion compression, $P_{o(1dB)}$**

output power where the conversion gain decreases by 1 dB compared with the linear conversion gain

3.14**noise figure, F**

see 702-08-57 of IEC 60050-702

NOTE The term "noise figure" expresses "noise factor" in decibels.

3.15**intermodulation distortion**

$$P_n/P_1$$

ratio of the n th order component of the output power to the fundamental component of the output power

NOTE The abbreviation " IMD_n " is in common use for the n th order intermodulation distortion.

[SOURCE: IEC 60747-4:2007/AMD1:2017, 7.2.19]

3.16**output power at the intercept point (for intermodulation products), $P_{n(IP)}$**

output power at the intersection between the extrapolated output powers of the fundamental component and the n th order intermodulation components, when the extrapolation is carried out in a diagram showing the output power of the components (in decibels) as a function of the input power (in decibels)

4 Abbreviated terms

The abbreviations used in this part of IEC 60747 are as follows:

RF Radio Frequency;
IF Intermediate Frequency;
LO Local Oscillator.

5 Essential ratings and characteristics**5.1 General**

This clause gives ratings and characteristics required for specifying integrated circuit microwave frequency converters.

5.1.1 Circuit identification and types**5.1.1.1 Designation and types**

The identification of type (device name), the category of circuit and technology applied shall be given.

Microwave frequency converters are divided into two categories:

- type A: down-converter;
- type B: up-converter.

5.1.1.2 General function description

A general description shall be made of the function performed by the integrated circuit microwave frequency converters and the features for the application.

5.1.1.3 Manufacturing technology

The manufacturing technology, for example, semiconductor monolithic integrated circuit, thin film integrated circuit, micro-assembly, shall be stated. This statement shall include details of the semiconductor technologies such as Schottky-barrier diode, MESFET, Si bipolar transistor, HBT.

5.1.1.4 Package identification

The following shall be stated:

- a) chip or packaged form;
- b) IEC and/or national reference number of the outline drawing, or drawing of non-standard package including terminal numbering;
- c) principal package material, for example, metal, ceramic, plastic;
- d) for chip form: outlines, dimensions, pad sizes, contact material, and recommended contact technologies.

5.1.1.5 Main application

The main application shall be stated if necessary. If the device has restrictive applications, these too shall be stated here.

5.2 Application description

Information on the application of the integrated circuit and its relation to the associated devices shall be given.

5.2.1 Conformance to system and/or interface information

It shall be stated whether the integrated circuit conforms to an application system and/or an interface standard or recommendation.

Detailed information about application systems, equipment and circuits such as VSAT systems, DBS receivers, microwave landing systems shall also be given.

5.2.2 Overall block diagram

A block diagram of the applied systems shall be given if necessary.

5.2.3 Reference data

The most important properties required to permit comparison between derivative types shall be given.

5.2.4 Electrical compatibility

It shall be stated whether the integrated circuit is electrically compatible with other particular integrated circuits or families of integrated circuits, or whether special interfaces are required.

Details shall be given concerning the type of the input and output circuits, for example, input/output impedances, d.c. block, open-drain.

Interchangeability with other devices, if any, shall be given.

5.2.5 Associated devices

If applicable, mention shall be made here of

- devices necessary for correct operation (list with type number, name, and function);
- peripheral devices with direct interfacing (list with type number, name, and function).

5.3 Specification of the function

5.3.1 Detailed block diagram – functional blocks

A detail block diagram or equivalent circuit information of the integrated circuit microwave frequency converters shall be given. The block diagram shall be composed of the following:

- a) functional blocks;
- b) mutual interconnections among the functional blocks;
- c) individual functional units within the functional blocks;
- d) mutual interconnections among the individual functional blocks;
- e) function of each external connection;
- f) interdependence between the separate functional blocks.

The block diagram shall identify the function of each external connection, and, where no ambiguity can arise, it can also show the terminal symbols and/or numbers. If the encapsulation has metallic parts, any connection to them from external terminals shall be indicated. The connections with any associated external electrical elements shall be stated, where necessary.

For the purpose of providing additional information, the complete electrical circuit diagram can be reproduced, though this will not necessarily involve giving indications of the function. Rules governing such diagrams may be obtained from IEC 60617.

5.3.2 Identification and function of terminals

All terminals shall be identified on the block diagram (supply terminals, input or output terminals, input/output terminals).

The terminal functions 1) to 4) shall be indicated, as shown in table 1 below.

Table 1 – Function of terminals

Terminal number	Terminal symbol	1 Terminal designation	2 Function	3 Input/output identification	4 Type of input/output circuits

5.3.2.1 Function 1: Terminal designation

The terminal designation to indicate the terminal function shall be given. Supply terminals, ground terminals, blank terminals (with abbreviation NC) and non-usable terminals (with abbreviation NU) shall be distinguished.

5.3.2.2 Function 2: Function

A brief indication of the terminal function shall be given:

- each function of multi-role terminals, i.e. terminals having multiple functions;
- each function of the integrated circuit selected by mutual pin connections, or programming and/or application of function selection data to the function selection pin such as mode selection pin.

5.3.2.3 Function 3: Input/output identification

Input, output, input/output, and multiples of the input/output terminal shall be distinguished.

5.3.2.4 Function 4: Type of input/output circuits

The type of input and output circuits, for example, input/output impedances, with or without d.c. block, etc., shall be distinguished.

5.3.2.5 Function 5: Type of ground

If the baseplate of the package is used as ground, this shall be stated.

EXAMPLE:

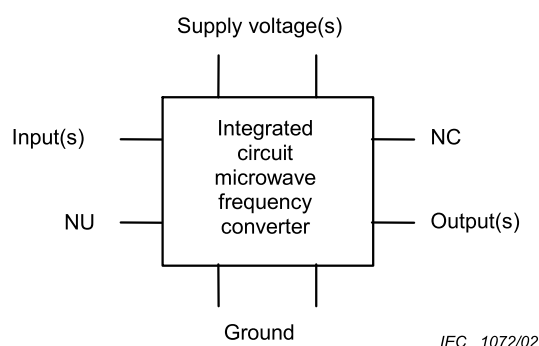


Figure 1 – Electrical terminal symbols

5.3.3 Function description

The function performed by the circuit shall be specified, including the following information:

- basic function;
- relation to external terminals;
- operation mode (e.g., set-up method, preference, etc.);
- interruption handling.

5.3.4 Family related characteristics

All family-specific functional descriptions shall be stated (with reference to IEC 60748-2, IEC 60748-3 and IEC 60748-4).

If ratings, characteristics and function characteristics exist for the family, the relevant part of IEC 60748 shall be used (for example, for microprocessors, see IEC 60748-2, chapter III, section 3).

NOTE For each new device family, specific items shall be added to the relevant part of IEC 60748.

5.4 Limiting values (absolute maximum rating system)

The table giving these values shall specify the following:

- any interdependence of limiting conditions;
- if externally connected and/or attached elements, for example heatsinks, have an influence on the values of the ratings, the ratings shall be prescribed for the integrated circuit with the elements connected and/or attached;
- if limiting values are exceeded for transient overload, the permissible excess and their durations shall be specified;
- where minimum and maximum values differ during programming of the device, this shall be stated;
- all voltages referenced to a specified reference terminal (V_{SS} , GND, etc.);
- in satisfying the following clauses, if maximum and/or minimum values are quoted, the manufacturer shall indicate whether he refers to the absolute magnitude or to the algebraic value of the quantity;
- the ratings given shall cover the operation of the multi-function integrated circuit over the specified range of operating temperatures. Where such ratings are temperature-dependent, such dependence shall be indicated.

5.4.1 Electrical limiting values

Limiting values shall be specified as follows:

Table 2 – Electrical limiting values

Subclause	Parameters	Min.	Max.
5.4.1.1	Power supply voltage(s)		+
5.4.1.2	Power supply current(s) (where appropriate)		+
5.4.1.3	Terminal voltage(s) (where appropriate)	+	+
5.4.1.4	Terminal current(s) (where appropriate)		+
5.4.1.5	Input power		+
5.4.1.6	LO input power		+
5.4.1.7	Power dissipation		+

The detail specification may indicate those values within table 2, including note 1 and note 2.

Parameters (see notes 1 and 2)	Symbols	Min.	Max.	Unit
NOTE 1 Where appropriate, in accordance with the type of considered circuit. NOTE 2 For power supply voltage range: limiting value(s) of the continuous voltage(s) at the supply terminal(s) with respect to a special electrical reference point; where appropriate, limiting value between specified supply terminals; when more than one voltage supply is required, a statement should be made as to whether the sequence in which these supplies are applied is significant: if so, the sequence should be stated; when more than one supply is needed, it may be necessary to state the combinations of ratings for these supply voltages and currents.				

5.4.2 Temperatures

- a) Operating temperature (ambient or reference-point temperature)
- b) Storage temperature
- c) Channel temperature
- d) Lead temperature (for soldering)

The detail specification may indicate those values within the table including the note.

Parameters (see note)	Symbols	Min.	Max.	Unit
NOTE Where appropriate, in accordance with the type of considered circuit.				

5.5 Operating conditions (within the specified operating temperature range)

These are not to be inspected, but may be used for quality assessment purposes.

5.5.1 Power supplies – Positive and/or negative values

5.5.2 Initialization sequences (where appropriate)

If special initialization sequences are necessary, the power supply sequencing and initialization procedure shall be specified.

5.6 Electrical characteristics

Each characteristic shall be stated: either

- a) over the specified range of operating temperatures, or
- b) at a temperature of 25 °C, and at maximum and minimum operating temperatures.

The parameters shall be specified corresponding to the type as shown in Table 3 below:

Table 3 – Electrical characteristics

Sub-clause	Parameters	Min.	Typical*	Max.	Types	
					A	B
5.6.1	Operating current		+	+	+	+
5.6.2	Conversion gain	+	+		+	+
5.6.3	Conversion gain flatness		+	+	+	+
5.6.4	LO/RF isolation	+	+		+	+
5.6.5	LO/IF isolation (where appropriate)	+	+		+	+
5.6.6	RF/IF isolation (where appropriate)	+	+		+	
5.6.7	IF/RF isolation (where appropriate)	+	+			+
5.6.8	Image rejection (where appropriate)	+	+		+	
5.6.9	Sideband suppression (where appropriate)	+	+			+
5.6.10	Output power at specified input power	+	+		+	+
5.6.11	Output power at 1 dB conversion compression	+	+		+	+
5.6.12	Noise figure		+	+	+	
5.6.13	Intermodulation distortion		+	+	+	+
5.6.14	Output power at the intercept point (for intermodulation products)	+	+		+	+
5.6.15	LO port return loss	+	+		+	+
5.6.16	RF port return loss	+	+		+	+
5.6.17	IF port return loss	+	+		+	+
* Optional.						
NOTE Conversion loss and conversion loss flatness can substitute for parameters 5.6.2 and 5.6.3.						

The detail specification may indicate these values within the table.

Characteristics	Symbols	Conditions	Min.	Typical*	Max.	Units
* Optional.						

5.7 Mechanical and environmental ratings, characteristics and data

Any specific mechanical and environmental ratings applicable shall be stated (see also 5.10 and 5.11 of IEC 60747-1).

5.8 Additional information

Where appropriate, the information detailed in the following subclauses shall be given.

5.8.1 Equivalent input and output circuit

Detail information shall be given regarding the type of input and output circuits, for example, input/output impedances, d.c. block, open-drain.

5.8.2 Internal protection

A statement shall be given to indicate whether the integrated circuit contains internal protection against high static voltages or electrical fields.

5.8.3 Capacitors at terminals

If capacitors for the input/output d.c. block are needed, these capacitances shall be stated.

5.8.4 Thermal resistance

5.8.5 Interconnections to other types of circuit

Where appropriate, details of the interconnections to other circuits shall be given.

5.8.6 Effects of externally connected component(s)

Curves or data indicating the effect of an externally connected component(s) that influence the characteristics may be given.

5.8.7 Recommendations for any associated device(s)

For example, decoupling of the power supply to a high frequency device shall be stated.

5.8.8 Handling precautions

Where appropriate, handling precautions specific to the circuit shall be stated (see also IEC 61340-5-1 and IEC 61340-5-2 concerning electrostatic-sensitive devices).

5.8.9 Application data**5.8.10 Other application information****5.8.11 Date of issue of the data sheet****6 Measuring methods****6.1 General**

This clause prescribes measuring methods for electrical characteristics of integrated circuit microwave frequency converters.

6.1.1 General precautions

The general precautions listed in 6.3, 6.4 and 6.6 of IEC 60747-1, shall be applied. In addition, special care shall be taken to use low-ripple d.c. supplies and to decouple adequately all bias supply voltages at the frequency of measurement. Special care concerning the load impedance of the test circuit shall also be taken to measure the output power.

The input signal (RF for down-converter and IF for up-converter) level shall be a small signal condition under which the devices exhibit linear characteristics against the input signal, unless otherwise specified.

6.1.2 Characteristic impedance

The input and output characteristic impedances of the measurement systems, shown in the circuit in this standard, are 50 Ω . If they are not 50 Ω , they shall be specified.

6.1.3 Handling precautions

When handling electrostatic-sensitive devices, the handling precautions given in IEC 61340-5-1 and IEC 61340-5-2, shall be observed.

6.1.4 Types

The devices in this standard are both package and chip types, measured using suitable test fixtures.

6.2 Conversion gain (G_c)**6.2.1 Purpose**

To measure conversion gain under specified conditions.

6.2.2 Circuit diagram

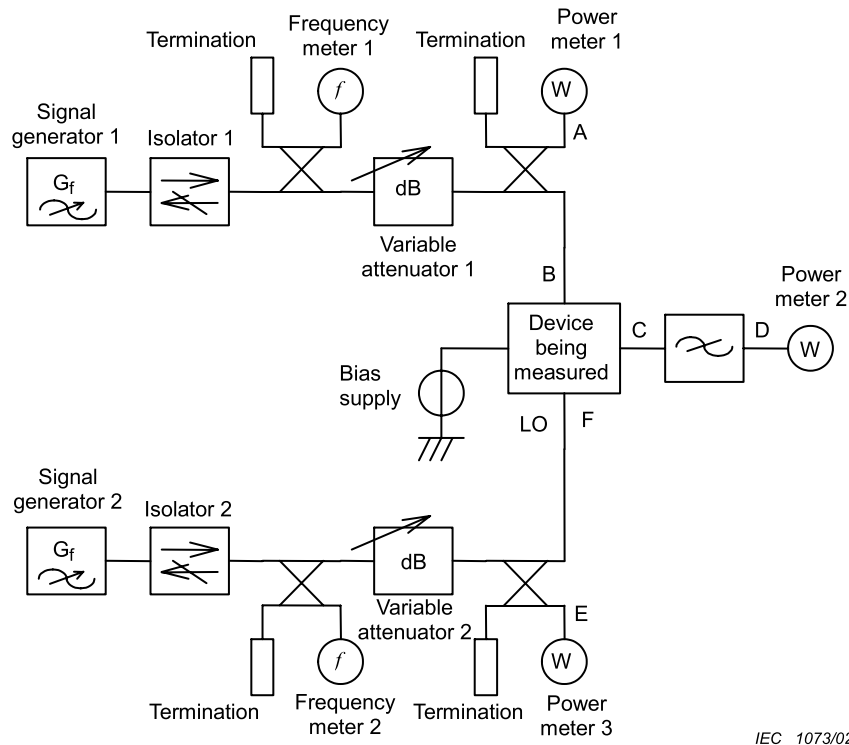


Figure 2 – Circuit diagram for the measurement of conversion gain

6.2.3 Principle of measurement

The conversion gain G_c , measured under the specified local oscillator power P_{LO} , is derived from the input power P_i and the output power P_o of the device, being measured as follows:

$$G_c = P_o - P_i \quad (1)$$

In the circuit diagram shown in Figure 2, P_i , P_o and P_{LO} are derived from the following equations:

$$P_i = P_1 - L_1 \quad (2)$$

$$P_o = P_2 + L_2 \quad (3)$$

$$P_{LO} = P_3 - L_3 \quad (4)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point C in dBm, minus the power at point D in dBm;

L_3 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1;

P_2 is the value indicated by power meter 2;

P_3 is the value indicated by power meter 3.

P_i , P_o , P_1 , P_2 , and P_3 are expressed in dBm. L_1 , L_2 , and L_3 are expressed in decibels (dB).

The conversion gain G_c is the power gain measured in the region where the change of output power in dBm is the same as that of the input power.

6.2.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatched at its input. The value of L_1 , L_2 , and L_3 shall be measured beforehand. The filter at the output rejects the undesired frequency band.

6.2.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during these measurements. Terminations shall be capable of handling the power in the test environment.

Harmonics or spurious responses of the signal generator shall be reduced to be negligible.

The conversion gain G_c shall be measured without the influence of filter impedance at LO and RF ports.

6.2.6 Measurement procedure

The frequency of the signal generator for the input signal shall be set to the specified value.

The frequency of the signal generator for the local signal shall be set to the specified value.

The power level of the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

The gain measured in this region is the conversion gain G_c .

6.2.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port
- Input frequency.

6.3 Conversion gain flatness (ΔG_c)

6.3.1 Purpose

To measure the conversion gain flatness under specified conditions.

6.3.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.3.3 Principle of measurement

See the principle of measurement of 6.2.3.

The linear gain flatness is derived from following equation

$$\Delta G_c = G_{c(max)} - G_{c(min)} \quad (5)$$

where $G_{c(max)}$ and $G_{c(min)}$ are the maximum and the minimum conversion gains in the specified frequency band at the specified input power, respectively.

There are two kinds of measurement methods of ΔG_c . One is the measurement under fixed local oscillator frequency and the other is the measurement under constant output frequency.

6.3.4 Circuit description and requirements

See the circuit description and requirements of 6.2.4.

6.3.5 Precaution to be observed

See the precaution to be observed of 6.2.5.

6.3.6 Measurement procedure

6.3.6.1 Conversion gain flatness for constant LO frequency

The frequency of the signal generator for the local signal shall be set to the specified value.

The frequency of the signal generator for the input signal shall be set to the specified value.

The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

Set the suitable input power level for measuring the conversion gain.

Vary the input frequency in the specified frequency band while keeping the input power level constant.

Obtain the maximum conversion gain $G_{c(max)}$ and the minimum conversion gain $G_{c(min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.6.2 Conversion gain flatness for constant output frequency

The frequency of the signal generator for the local signal shall be set to obtain the specified output frequency.

The frequency of the signal generator for the input signal within the specified input frequency range shall be set to obtain the specified output frequency.

The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

Set the suitable input power level for measuring the conversion gain.

Vary the input and local frequencies within the specified frequency band while keeping the input power level and the local power level constant.

Obtain the maximum conversion gain $G_{c(max)}$ and the minimum conversion gain $G_{c(min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.6.3 Conversion gain flatness for constant input frequency

The frequency of the signal generator for the local signal shall be set to the specified value.

The frequency of the signal generator for the input signal shall be set to the specified value.

The bias under specified conditions is applied.

The power level of the local signal shall be adjusted to the specified value.

An adequate input power shall be applied to the device being measured.

By varying the input power, confirm that a change of output power corresponds to an equal change in the input power.

Set the suitable input power level for measuring the conversion gain.

Vary the local frequency within the specified frequency band while keeping the local power level constant.

Obtain the maximum conversion gain $G_{c(max)}$ and the minimum conversion gain $G_{c(min)}$ in the specified frequency band.

The conversion gain flatness ΔG_c is calculated using equation (5).

6.3.7 Specified conditions

6.3.7.1 Conversion gain flatness for constant LO frequency

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port
- Input frequency band.

6.3.7.2 Conversion gain flatness for constant output frequency

- Ambient or reference point temperature
- Bias conditions
- Frequency band of local signal
- Incident power of local port

- Input frequency range
- Output frequency.

6.3.7.3 Conversion gain flatness for constant input frequency

- Ambient or reference point temperature
- Bias conditions
- Frequency band of local signal
- Incident power of local port
- Input frequency

6.4 LO/IF isolation ($P_{LO}/P_{LO(IF)}$)

6.4.1 Purpose

To measure the isolation from the local port to the IF port under specified conditions.

6.4.2 Circuit diagram

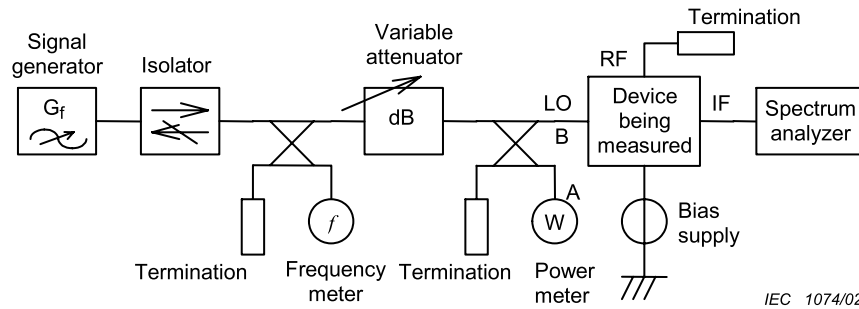


Figure 3 – Circuit diagram for the measurement of the LO/IF isolation

6.4.3 Principle of measurement

In the circuit diagram shown in Figure 3, the input local power P_{LO} in dBm is derived as follows:

$$P_{LO} = P_1 - L_1 \quad (6)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

P_1 is the value indicated by the power meter in dBm.

$P_{LO(IF)}$ is the local frequency component of the output power at the IF port in dBm when the specified P_{LO} in dBm is applied.

The local port to IF port isolation $P_{LO}/P_{LO(IF)}$ in decibels (dB) is derived as follows:

$$P_{LO}/P_{LO(IF)} = P_{LO} - P_{LO(IF)} \quad (7)$$

where $P_{LO(IF)}$ is the value indicated by the spectrum analyzer.

6.4.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its local ports.

L_1 shall be measured beforehand.

NOTE The spectrum analyzer can be replaced with a power meter and a filter having proper frequency response and impedance characteristics. In the case of such a set-up, the filter must reject the undesired frequency components, and should not change the terminal impedances for frequencies such as RF, LO and IF.

6.4.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during the measurement.

Harmonics or spurious responses of the signal generator shall be reduced so as to be negligible.

6.4.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} , which is given by equation (6), is applied.

The power level of the local frequency component at the IF port $P_{LO(IF)}$ is measured by the spectrum analyzer.

The local port to IF port isolation $P_{LO} / P_{LO(IF)}$ is calculated from equation (7).

6.4.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port.

6.5 LO/RF isolation ($P_{LO} / P_{LO(RF)}$)

6.5.1 Purpose

To measure the isolation from the local port to the RF port under specified conditions.

6.5.2 Circuit diagram

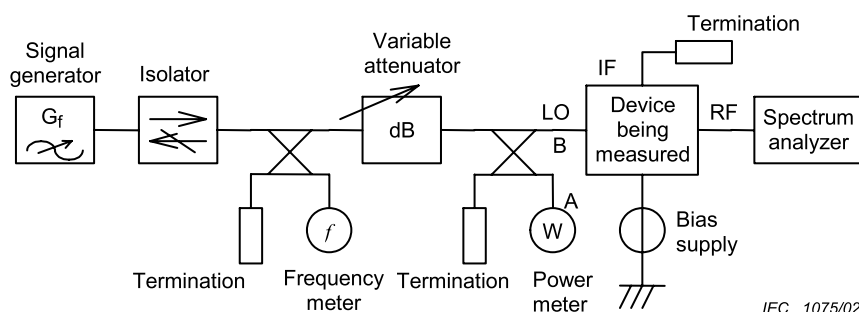


Figure 4 – Circuit diagram for the measurement of the LO/RF isolation

6.5.3 Principle of measurement

In the circuit diagram shown in Figure 4, the input local power P_{LO} in dBm is derived as follows:

$$P_{LO} = P_1 - L_1 \quad (8)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

P_1 is the value indicated by the power meter in dBm.

$P_{LO(RF)}$ is the local frequency component of the output power at the RF port in dBm when the specified P_{LO} in dBm is applied.

The local port to RF port isolation $P_{LO}/P_{LO(RF)}$ in decibels (dB) is derived as follows:

$$P_{LO}/P_{LO(RF)} = P_{LO} - P_{LO(RF)} \quad (9)$$

where $P_{LO(RF)}$ is the value indicated by the spectrum analyzer.

6.5.4 Circuit description and requirements

See the circuit description and requirements described in 6.4.4.

6.5.5 Precautions to be observed

See the precautions to be observed described in 6.4.5.

6.5.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power, P_{LO} , which is given by equation (8), is applied.

The power level of the local frequency component at the RF port $P_{LO(RF)}$ is measured by the spectrum analyzer.

The local port to RF port isolation $P_{LO}/P_{LO(RF)}$ is calculated from equation (9).

6.5.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of local signal
- Incident power of local port
- Termination condition at IF port.

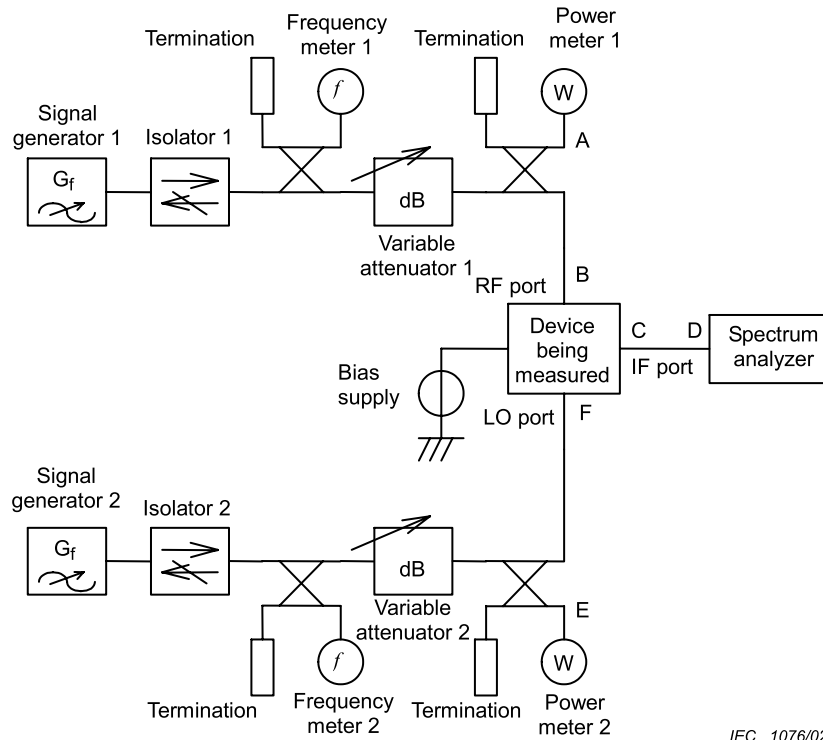
6.6 RF/IF, IF/RF isolation

6.6.1 RF/IF isolation ($P_{RF}/P_{RF(IF)}$) – Type A

6.6.1.1 Purpose

To measure the isolation from the RF port to the IF port under specified conditions.

6.6.1.2 Circuit diagram



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Figure 5 – Circuit diagram for the measurement of the RF/IF isolation for type A

6.6.1.3 Principle of measurement

In the circuit diagram shown in Figure 5, the input RF power P_{RF} and the LO power P_{LO} are derived as follows:

$$P_{RF} = P_1 - L_1 \quad (10)$$

$$P_{LO} = P_2 - L_2 \quad (11)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 at the RF port in dBm;

P_2 is the value indicated by power meter 2 at the LO port in dBm.

$P_{RF(IF)}$ is the RF frequency component of the output power at the IF port in dBm when the specified P_{RF} and P_{LO} are applied.

The isolation $P_{RF}/P_{RF(IF)}$ from the RF port to the IF port is derived as follows:

$$P_{RF}/P_{RF(IF)} = P_{RF} - P_{RF(IF)} \quad (12)$$

where $P_{RF(IF)}$ is the value indicated by the spectrum analyzer in dBm.

6.6.1.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its RF and LO ports.

L_1 and L_2 shall be measured beforehand.

6.6.1.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during measurement.

Harmonics or spurious responses of the signal generators shall be reduced so as to be negligible.

6.6.1.6 Measurement procedure

The frequency of the signal generator 1 for the RF input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The local power P_{LO} , which is given by equation (11), is applied and adjusted to the specified value by the variable attenuator 2.

The RF power P_{RF} , which is given by equation (10), is applied and adjusted to the specified value by the variable attenuator 1.

The power level of the RF signal component at the IF port $P_{RF(IF)}$ is measured by the spectrum analyzer.

The isolation $P_{RF}/P_{RF(IF)}$ from the RF port to the IF port is calculated from the equation (12).

6.6.1.7 Specified conditions

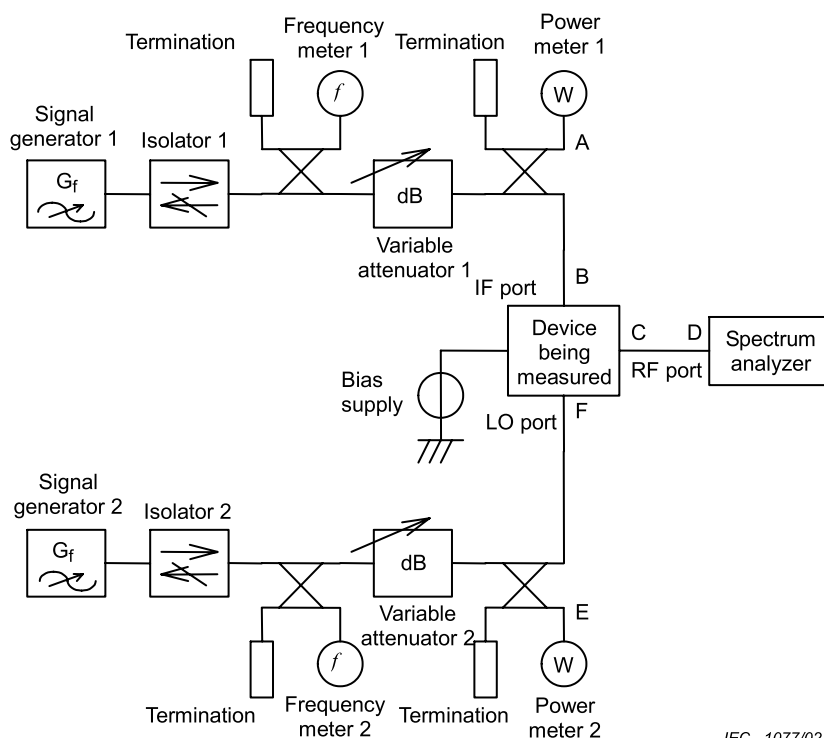
- Ambient or reference point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Input power of RF port
- Frequency of RF signal.

6.6.2 IF/RF isolation ($P_{IF}/P_{IF(RF)}$) – Type B

6.6.2.1 Purpose

To measure the isolation from the IF port to the RF port under specified conditions.

6.6.2.2 Circuit diagram



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Figure 6 – Circuit diagram for the measurement of the IF/RF isolation for type B

6.6.2.3 Principle of measurement

In the circuit diagram shown in Figure 6, the input IF power P_{IF} and the LO power P_{LO} are derived as follows:

$$P_{IF} = P_1 - L_1 \quad (13)$$

$$P_{LO} = P_2 - L_2 \quad (14)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 at the IF port in dBm;

P_2 is the value indicated by power meter 2 at the LO port in dBm.

$P_{IF(RF)}$ is the IF frequency component of the output power at the RF port in dBm when the specified P_{IF} and P_{LO} are applied.

The isolation $P_{IF} / P_{IF(RF)}$ from the IF port to the RF port is derived as follows:

$$P_{IF} / P_{IF(RF)} = P_{IF} - P_{IF(RF)} \quad (15)$$

where $P_{IF(RF)}$ is the value indicated by the spectrum analyzer in dBm.

6.6.2.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its IF and LO port.

L_1 and L_2 shall be measured beforehand.

6.6.2.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during measurement.

Harmonics or spurious responses of the signal generators shall be reduced to be negligible.

6.6.2.6 Measurement procedure

The frequency of the signal generator 1 for the IF input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The local power P_{LO} , which is given by equation (14), is applied and adjusted to the specified value by the variable attenuator 2.

The IF power P_{IF} , which is given by equation (13), is applied and adjusted to the specified value by the variable attenuator 1.

The power level of the IF signal component at the RF port $P_{IF(RF)}$ is measured by the spectrum analyzer.

The isolation $P_{IF}/P_{IF(RF)}$ from the IF port to the RF port is calculated from equation (15).

6.6.2.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Input power of IF port
- Frequency of IF signal.

6.7 Image rejection ($P_o/P_{o(im)}$)

6.7.1 Purpose

To measure the image rejection under specified conditions.

6.7.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.7.3 Principle of measurement

The image frequency $f_{(\text{image})}$ is defined as follows:

$$f_{(\text{image})} = 2f_{(\text{local})} - f_{(\text{rf})} \quad (16)$$

where

$f_{(\text{local})}$ is the specified local frequency;

$f_{(\text{rf})}$ is the specified RF frequency.

The input power P_i and the local power P_{LO} are as follows:

$$P_i = P_1 - L_1 \quad (17)$$

$$P_{\text{LO}} = P_3 - L_3 \quad (18)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_3 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 in dBm;

P_3 is the value indicated by power meter 3 in dBm.

P_i , P_{LO} , P_1 and P_3 are expressed in dBm. L_1 and L_3 are expressed in decibels (dB).

The image rejection $P_o/P_{o(\text{im})}$ in decibels (dB) is derived as follows:

$$P_o / P_{o(\text{im})} = P_{o(\text{rf})} - P_{o(\text{image})} \quad (19)$$

where

$P_{o(\text{rf})}$ is the value indicated by power meter 2 in dBm when the input frequency and power are the specified RF frequency and input power;

$P_{o(\text{image})}$ is the value indicated by power meter 2 in dBm when the input frequency and power are the image frequency and input power.

6.7.4 Circuit description and requirements

See the circuit description and requirements of 6.2.4.

6.7.5 Precaution to be observed

See the precaution to be observed of 6.2.5.

6.7.6 Measurements procedure

The frequency of the signal generator for the input signal shall be adjusted to the specified RF frequency.

The power level of the input signal (P_i) shall be adjusted to the specified input power.

The frequency of the signal generator for the local signal shall be adjusted to the specified local frequency.

The power level of the local signal (P_{LO}) shall be adjusted to the specified local power.

The bias under specified conditions is applied to the device being measured.

The output power ($P_{o(rf)}$) is measured by power meter 2.

The frequency of the signal generator for the input signal shall be changed to the specified image frequency.

The power level of the input signal (P_i) shall be adjusted to the specified input power.

The output power ($P_{o(image)}$) is measured by power meter 2.

The image rejection is calculated from equation (19).

6.7.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- RF frequency
- Local frequency
- Input power of RF port
- Local power.

6.8 Sideband suppression ($P_o / P_{o(U)}$)

6.8.1 Purpose

To measure the sideband suppression under specified conditions.

6.8.2 Circuit diagram

See the circuit diagram shown in Figure 6.

6.8.3 Principle of measurement

The incident power of the IF port P_{IF} and the local power P_{LO} are derived as follows:

$$P_{IF} = P_1 - L_1 \quad (20)$$

$$P_{LO} = P_2 - L_2 \quad (21)$$

where

L_1 is the power at point A in dBm, minus the power at point B in dBm;

L_2 is the power at point E in dBm, minus the power at point F in dBm;

P_1 is the value indicated by power meter 1 in dBm;

P_2 is the value indicated by power meter 2 in dBm.

The sideband suppression $P_o / P_{o(U)}$ is derived as follows:

$$P_o / P_{o(U)} = P_o - P_{o(U)} \quad (22)$$

In the circuit diagram shown in Figure 6, the output power of the desired sideband P_o and the output power of the undesired sideband $P_{o(U)}$ are derived from the following equations:

$$P_o = P_3 + L_3 \quad (23)$$

$$P_{o(U)} = P_4 + L_4 \quad (24)$$

where

- L_3 is the power at point C for the desired sideband frequency in dBm, minus the power at point D for the desired sideband frequency in dBm;
- L_4 is the power at point C for the undesired sideband frequency in dBm, minus the power at point D for the undesired sideband frequency in dBm;
- P_3 is the value for the desired sideband frequency indicated by the spectrum analyzer in dBm;
- P_4 is the value for the undesired sideband frequency indicated by the spectrum analyzer in dBm.

6.8.4 Circuit description and requirements

See the circuit description and requirements of 6.6.2.4. The circuit losses L_1 , L_2 , L_3 and L_4 shall be measured beforehand.

6.8.5 Precautions to be observed

See the precautions to be observed in 6.6.2.5.

6.8.6 Measurement procedure

The IF frequency of the signal generator 1 shall be adjusted to the specified value.

The local frequency of the signal generator 2 shall be adjusted to the specified value.

The bias under specified conditions is applied to the device being measured.

The incident power of the IF port P_{IF} shall be adjusted to the specified value.

The local power P_{LO} shall be adjusted to the specified value.

The value of P_3 is measured with the spectrum analyzer.

The value of P_4 is measured with the spectrum analyzer.

The sideband suppression $P_o/P_{o(U)}$ is calculated from equations (22), (23) and (24).

6.8.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Incident power of IF port
- IF frequency
- Local power
- Local frequency.

6.9 Output power (P_o)

6.9.1 Purpose

To measure the output power under specified conditions.

6.9.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.9.3 Principle of measurement

In the circuit diagram shown in Figure 1, the output power P_o under the specified local signal and the input signal is derived from equation (3).

6.9.4 Circuit description and requirements

See the circuit description and requirements described in 6.2.4.

6.9.5 Precautions to be observed

See the precautions to be observed described in 6.2.5.

6.9.6 Measurement procedure

The frequency of the signal generator 1 for the input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} is applied.

The specified input power P_i is applied.

The power level P_2 is measured.

The output power is calculated from equation (3).

6.9.7 Specified conditions

- Ambient or reference point temperature
- Bias conditions
- Frequency of input signal
- Power level of input signal
- Local power
- Frequency of local signal.

6.10 Output power at 1-dB conversion compression ($P_{o(1dB)}$)

6.10.1 Purpose

To measure the output power at 1-dB conversion compression under specified conditions.

6.10.2 Circuit diagram

See the circuit diagram shown in Figure 2.

6.10.3 Principle of measurement

See the principle of measurement in 6.2.3.

The output power at 1-dB conversion compression $P_{o(1dB)}$ is the output power where the conversion gain decreases by 1 dB compared with G_c .

6.10.4 Circuit description and requirements

See the circuit description and requirements described in 6.2.4.

6.10.5 Precautions to be observed

See the precautions to be observed described in 6.2.5.

6.10.6 Measurement procedure

The frequency of the signal generator 1 for the input signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the local signal shall be adjusted to the specified value.

The bias under specified conditions is applied.

The specified local power P_{LO} is applied.

The adequate input power is applied.

By varying the input power, confirm the change of conversion gain in decibels is the same as that of input power.

The input power is increased till the conversion gain decreases by 1 dB, compared with G_c .

The power level P_2 is measured.

The output power at 1-dB conversion compression $P_{o(1dB)}$ is calculated from equation (3).

6.10.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of input signal
- Local power
- Frequency of local signal.

6.11 Noise figure (F)

6.11.1 Purpose

To measure the noise figure under specified conditions.

6.11.2 Circuit diagram

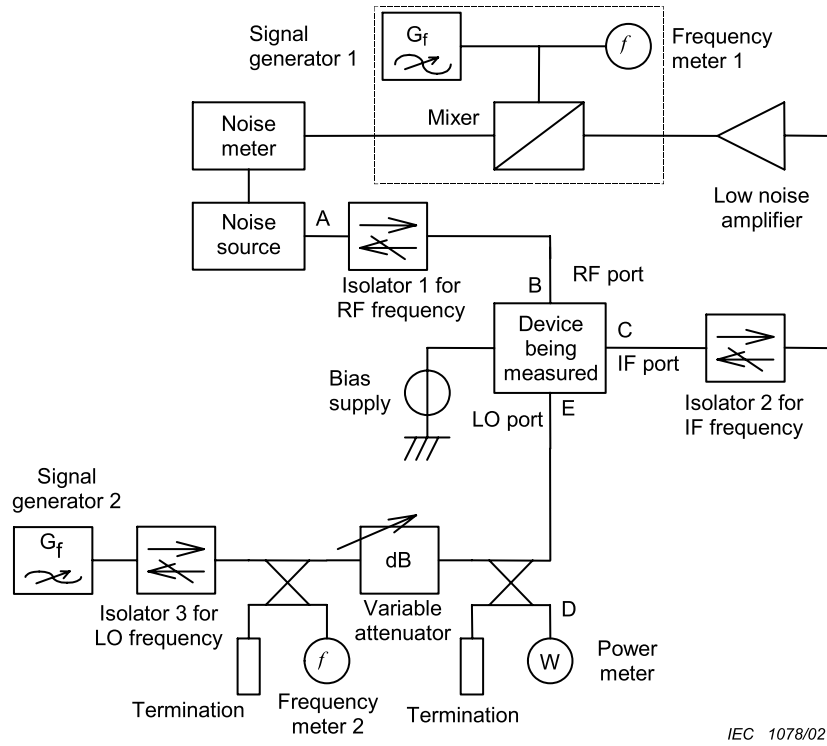


Figure 7 – Circuit diagram for measurement of noise figure

6.11.3 Principle of measurement

The noise figure F of the device being measured is derived from the following equation, when the oscillation power P_{LO} is applied to the device being measured:

$$F = 10 \log \left(10^{(F_{12} - L_1) / 10} - \frac{10^{(F_2 / 10)} - 1}{10^{(G_c / 10)}} \right) \quad (25)$$

The noise figure measurement is carried out by using the hot and cold measurement method; F_{12} , F_2 , and G_c are calculated as follows:

$$F_{12} = 10 \log \left(\frac{10^{(ENR_1 / 10)}}{(P_{N1} / P_{N2}) - 1} \right) \quad (26)$$

$$F_2 = 10 \log \left(\frac{10^{(ENR_2 / 10)}}{(P_{N3} / P_{N4}) - 1} \right) \quad (27)$$

$$G_c = 10 \log \left(\frac{P_{N1} - P_{N2}}{P_{N3} - P_{N4}} \right) + (ENR_1 - ENR_2) + L_1 \quad (28)$$

The input local power P_{LO} is derived as follows:

$$P_{LO} = P_1 - L_2 \quad (29)$$

where

- F_{12} is the overall noise figure in decibels (dB);
- F_2 is the noise figure in decibels (dB) after point C;
- ENR_1 is the excess noise ratio of the noise source at the RF frequency;
- ENR_2 is the excess noise ratio of the noise source at the IF frequency;
- P_{N1} is the measured noise powers in watts (W) under the hot state of the noise source;
- P_{N2} is the measured noise powers in watts (W) under the cold state of the noise source;
- P_{N3} is the measured noise powers in watts (W) under the hot state of the noise source in the case of directly connecting point A to C;
- P_{N4} is the measured noise powers in watts (W) under the cold state of the noise source in the case of directly connecting point A to C;
- P_1 is the value indicated by the power meter at the LO port in decibels (dB);
- L_1 is the circuit loss in decibels (dB) between point A and B;
- L_2 is the power at point D in decibels (dB), minus the power at point E in decibels (dB).

The temperature of the measurement is 290 K.

6.11.4 Circuit description and requirements

The purpose of isolator 1 and isolator 3 is to enable the power level to the device being measured to be kept constant, irrespective of impedance mismatches at its RF and LO ports.

The purpose of isolator 2 is to enable the power level to the low noise amplifier to be kept constant, irrespective of impedance mismatches.

When the IF frequency is within the noise meter operating range, the circuit surrounded by the dotted line can be omitted.

When the RF frequency and the IF frequency are not covered by the one noise source frequency range, the noise source for the IF frequency and the noise source for the RF frequency are needed.

The circuit loss L_1 and L_2 shall be measured beforehand.

6.11.5 Precautions to be observed

The entire circuit shall be shielded and earthed to prevent undesired signals. For noise figure measurement under the SSB condition, careful attention shall be paid to the image and to other spurious responses that are generated by the mixer and device being measured.

These spurious responses shall be reduced so as to be negligible.

6.11.6 Measurement procedure

The frequency of the signal generator 1 is adjusted to meet the specified condition.

NOTE The measurement procedure is omitted when the mixer circuit is not used.

In order to measure the noise contribution of the measurement system, connect point A to C in Figure 7 without the device being measured.

The noise powers P_{N3} and P_{N4} corresponding to the hot and cold noise source, respectively, are measured.

The device being measured is inserted as shown in Figure 7.

The frequency of the signal generator 2 is adjusted to meet the specified condition.

The bias under specified conditions is applied.

The P_{LO} under specified conditions are applied.

The noise powers P_{N1} and P_{N2} corresponding to the hot and cold noise source, respectively, are measured.

The noise figure in decibels (dB) is calculated by equation (25).

6.11.7 Specified conditions

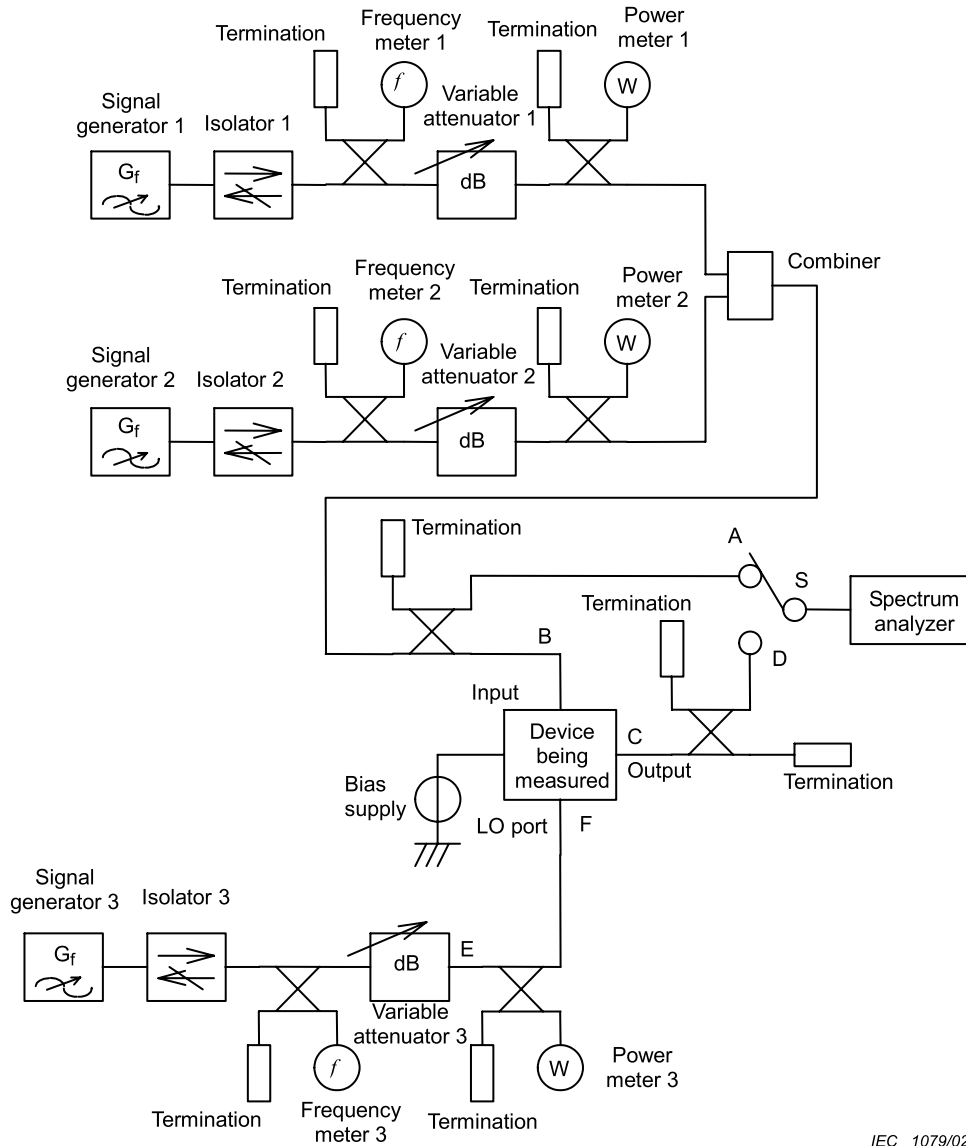
- Ambient or reference-point temperature
- Bias conditions
- Local power
- Frequency of local signal
- Single-side band or double-side band.

6.12 Intermodulation distortion (P_n/P_1)

6.12.1 Purpose

To measure the intermodulation distortion under specified conditions.

6.12.2 Circuit diagram



IEC 1079/02

Figure 8 – Circuit for the measurement of intermodulation distortion

6.12.3 Principle of measurement

In the circuit diagram shown in Figure 8, the input power P_i , the output power P_o and the intermodulation products output power P_n , the local input power P_{LO} of the device being measured, are defined from the following equations:

$$P_i = P_1 + L_1 \quad (30)$$

$$P_o = P_2 + L_2 \quad (31)$$

$$P_n = P_3 + L_2 \quad (32)$$

$$P_{LO} = P_4 - L_3 \quad (33)$$

P_1 , P_2 and P_3 are the values indicated by the spectrum analyzer and correspond to P_i , P_o and P_n , respectively. P_4 is the output power at point E calculated from the value indicated by signal generator 3.

L_1 , L_2 and L_3 are the circuit losses from point A to point B, from point C to point D and from point E to point F shown in Figure 8, respectively.

P_i , P_o , P_n and P_{LO} are expressed in dBm. L_1 , L_2 and L_3 are expressed in decibels (dB).

The intermodulation distortion (P_n/P_1) in dBc is derived from equations (31) and (32) as follows:

$$P_n/P_1 = P_n - P_o = P_3 - P_2 \quad (34)$$

6.12.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its ports.

L_1 shall be measured beforehand.

The input and output ports correspond to the RF and IF ports for type A devices, and to the IF and RF ports for type B devices, respectively.

6.12.5 Precautions to be observed

See the precautions to be observed in 6.4.5.

It is better to terminate port D when the switch is connected to position A, and vice versa.

6.12.6 Measurement procedure

The bias under specified conditions is applied.

The signal generator 3 is tuned on and the LO signal is added to the device being measured with the specified level P_4 using the variable attenuator 3.

The switch is connected to position A.

The signal generator 1 is tuned on and the input signal is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 1.

The signal generator 2 is tuned on and the input signal is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 2.

The switch is connected to position D.

The output power P_2 and the intermodulation products P_3 in dBm are measured using the spectrum analyzer.

The intermodulation distortion on the specified input power P_1 is derived from equations (30) to (34).

6.12.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Input power
- Local power
- Input frequency
- Local frequency
- n th order.

6.13 Output power at the intercept point (for intermodulation products) ($P_{n(IP)}$)

6.13.1 Purpose

To measure the output power at the intercept point (for intermodulation products) under specified conditions.

6.13.2 Circuit diagram

See the circuit diagram of 6.12.2.

6.13.3 Principle of measurement

Refer to the principle of measurement of 6.12.3.

6.13.4 Circuit description and requirements

See the circuit description and requirements of 6.12.4.

6.13.5 Precautions to be observed

See the precautions to be observed in 6.12.5.

6.13.6 Measurement procedure

The bias under specified conditions is applied.

The signal generator 3 is tuned on and the LO signal is added to the device being measured with the specified level P_4 using the variable attenuator 3.

The switch is connected to position A.

The signal generator 1 is tuned on and the input power is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 1.

The signal generator 2 is tuned on and the input power is added to the device being measured with the specified level P_1 using the spectrum analyzer and the variable attenuator 2.

The switch is connected to position D.

The output power P_2 and the intermodulation products P_3 in dBm are measured using the spectrum analyzer.

Changing the power level of the input power using the variable attenuators 1 and 2, the above procedure is repeated within the specified range.

The obtained data are plotted on linear scales.

The straight lines of the output power and intermodulation products in the linear region are extended (slope of 1 and 3 are respectively observed).

The output power where the two extended lines intercept is the output power at the intercept point (for intermodulation products) under specified conditions.

6.13.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Input power range
- Local power
- Input frequency
- Local frequency
- n th order.

6.14 LO port return loss ($L_{\text{ret(LO)}}$)

6.14.1 Purpose

To measure the LO port return loss under specified conditions.

6.14.2 Circuit diagram

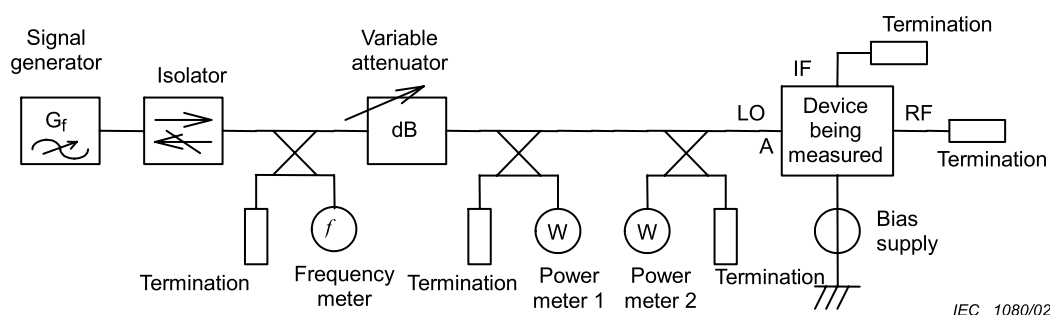


Figure 9 – Circuit for the measurement of the LO port return loss

6.14.3 Principle of measurement

In the circuit diagram shown in Figure 9, the LO port return loss is derived from the following equation:

$$L_{\text{ret(LO)}} = P_1 - P_2 \quad (35)$$

where

- P_1 is the value indicated by power meter 2 in dBm, when the incident LO power is totally reflected at point A;
- P_2 is the value indicated by power meter 2 in dBm, when the LO port of the device being measured is connected to point A.

6.14.4 Circuit description and requirements

The purpose of the isolator is to enable the power level to the device being measured to be kept constant, irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and other ports are connected to the termination.

The directivity of the directional coupler is enough compared to the return loss of the device being measured.

6.14.5 Precautions to be observed

Oscillation, which is checked by the spectrum analyzer, shall be eliminated during these measurements. The terminations shall be capable of handling the power in the test environment.

Harmonics or spurious responses of the signal generator shall be reduced to be negligible.

6.14.6 Measurement procedure

The frequency of the signal generator shall be adjusted to the specified value.

Point A is shunted.

Then, the power of power meter 2 is measured. This power is P_1 .

The LO port of the device being measured is then connected with point A and the other ports are connected to the termination.

The bias under specified conditions is applied.

Then, the power of power meter 2 is measured. This power is P_2 .

The LO port return loss is derived from equation (35).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.14.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of LO signal
- Incident power at the LO port.

6.15 RF port return loss ($L_{\text{ret(RF)}}$)

6.15.1 Purpose

To measure the RF port return loss under specified conditions.

6.15.2 Circuit diagram

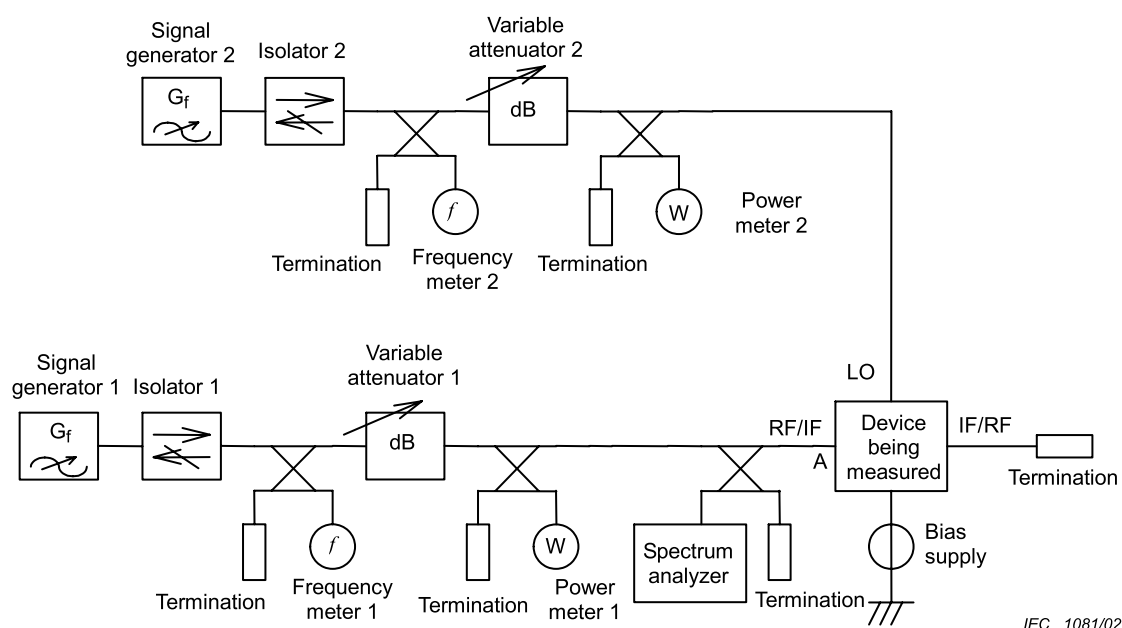


Figure 10 – Circuit for the measurement of the RF/IF port return loss

6.15.3 Principle of measurement

In the circuit diagram shown in Figure 10, the RF port return loss is derived from the following equation:

$$L_{\text{ret(RF)}} = P_1 - P_2 \quad (36)$$

where

P_1 is the value indicated by the spectrum analyzer in dBm, when the incident RF power is totally reflected at point A;

P_2 is the value indicated by the spectrum analyzer in dBm, when the RF port of the device being measured is connected to point A.

6.15.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and the other ports are connected to the terminations.

The directivity of the directional coupler is sufficient compared to the return loss of the device being measured.

6.15.5 Precautions to be observed

See the precautions to be observed in 6.2.5.

6.15.6 Measurement procedure

The frequency of the signal generator 1 for the RF signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the LO signal shall be adjusted to the specified value.

The power level of the LO signal shall be adjusted to the specified value.

Point A is shunted.

Then, the power of the spectrum analyzer is measured. This power is P_1 .

The RF port of the device being measured is then connected with point A.

The bias under specified conditions is applied.

Then, the power of the spectrum analyzer is measured. This power is P_2 .

The RF port return loss is derived from equation (36).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.15.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of RF signal
- Incident power at the RF port
- Frequency of LO signal
- Incident power at the LO port.

6.16 IF port return loss ($L_{\text{ret(IF)}}$)**6.16.1 Purpose**

To measure the IF port return loss under specified conditions.

6.16.2 Circuit diagram

See the circuit diagram shown in Figure 10.

6.16.3 Principle of measurement

In the circuit diagram shown in Figure 10, the IF port return loss is derived from following equation:

$$L_{\text{ret(IF)}} = P_1 - P_2 \quad (37)$$

where

P_1 is the value indicated by the spectrum analyzer in dBm, when the incident IF power is totally reflected at point A;

P_2 is the value indicated by the spectrum analyzer in dBm, when the IF port of the device being measured is connected to point A.

6.16.4 Circuit description and requirements

The purpose of the isolators is to enable the power level to the device being measured to be kept constant irrespective of impedance mismatches at its inputs.

The input port of the device being measured is connected to the directional coupler and other ports are connected to the terminations.

The directivity of the directional coupler is sufficient compared to the return loss of the device being measured.

6.16.5 Precautions to be observed

See the precautions to be observed in 6.2.5.

6.16.6 Measurement procedure

The frequency of the signal generator 1 for the IF signal shall be adjusted to the specified value.

The frequency of the signal generator 2 for the LO signal shall be adjusted to the specified value.

The power level of the LO signal shall be adjusted to the specified value.

Point A is shunted.

The power of the spectrum analyzer is then measured. This power is P_1 .

Then, the IF port of the device being measured is connected with point A.

The bias under specified conditions is applied.

The power of the spectrum analyzer is then measured. This power is P_2 .

The IF port return loss is derived from equation (37).

NOTE When measuring the total reflection power, point A can be opened if it is difficult to shunt it.

6.16.7 Specified conditions

- Ambient or reference-point temperature
- Bias conditions
- Frequency of IF signal
- Incident power at the IF port
- Frequency of LO signal
- Incident power at the LO port.

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IEC 60747-4:2007/AMD 1:2017

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